AURALEARN: AI-POWERED ADAPTIVE LEARNING PLATFORM

PROJECT REPORT

Submitted by

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of
Master of Computer Applications

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DECLARATION

I undersigned hereby declare that the project report ("**AuraLearn: AI-Powered Adaptive Learning Platform**"), submitted for partial fulfillment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Mr. AJAYAKUMAR KK**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **"AuraLearn: AI-Powered Adaptive Learning Platform"** submitted by **Muhammed Yaseen K (KMC24MCA-2025)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications is a bonafide record of the project work carried out by him under my supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Internal Supervisor

Project Coordinator

Head of the Department

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ABSTRACT

This Project aims to develop AuraLearn, a Personalized adaptive learning Platform designed to enhance student mastery of curriculum-based subjects. Leveraging the advanced capabilities of Gemini 2.5 flash, AuraLearn will Provide intelligent, context aware tutoring and dynamic learning Path adjustments. The Platform will feature 3 key user roles: Students, Knowledge Providers, and Admins. Students will benefit from personalized schedules generated with AI assistance based on their initial performance and availability, along with an AI tutor that delivers grounded explanations, adaptive quizzes, and remedial guidance, all Powered by Gemini's real-time conversational and structured output capabilities. knowledge Providers will contribute and refine educational content via AI assisted authoring and structured review workflows. Admin will oversee the curriculum, manage user access and maintain the system's foundational knowledge graph, which explicitly defines syllabus topics, granular content chunks, and their Prerequisite relationships.

Contents

ACKNOWLEDGEMENT i

ABSTRACT ii

CONTENTS iii

LIST OF TABLES vi

LIST OF FIGURES vii

ABBREVIATIONS viii

CHAPTER 1 Introduction 1

1.1 General Background 1

1.2 Objective 2

CHAPTER 2 Literature Survey 3

2.1 The Challenge of Personalized Learning 3

2.2 Paradigms in Adaptive Educational Systems 3

2.3 AI in Content and Assessment Generation 4

2.4 Synthesis and Research Gap 5

CHAPTER 3 System Analysis 6

3.1 Existing System 6

3.2 Proposed System 7

3.3 Module Description 9

3.4 Feasibility Study 10

3.4.1 Operational Feasibility 10

3.4.2 Technical Feasibility 11

3.4.3 Economic Feasibility 11
3.5 System Environment 12
3.6 Actors and Their Roles 13
CHAPTER 4 System Design 14
4.1 Methodology 14
4.2 User Story 15
4.3 Product Backlog 17
4.4 Project Plan 19
4.5 UML Diagrams 20
4.5.1 Use Case Diagram 20
4.5.2 Activity Diagram 21
4.5.3 Class Diagram 22
4.6 Database Design 23
4.7 User Interface Design 28
CHAPTER 5 TESTING AND IMPLEMENTATION 45
5.1 Implementation 45
5.2 Testing 46
5.2.1 Test Case 1 46
5.2.2 Test Case 2 47
5.2.3 Test Case 3 48
5.2.4 Test Case 4 49
CHAPTER 6 RESULTS AND DISCUSSION 51
6.1 System Functionality and Performance 51
6.2 Evaluation of AI-Powered Features 52
6.3 Challenges and Limitations 52
6.4 Achievement of Objectives 53
CHAPTER 7 CONCLUSION AND FUTURE WORK 54
7.1 Conclusion 54

7.2 Future Work 54

REFERENCES 56

APPENDICES 57

LIST OF TABLES

4.1 AuraLearn User Stories 16

4.2 AuraLearn Product Backlog 18

4.3 AuraLearn Project Plan 19

4.4 users Collection 24

4.5 subjects Collection 25

4.6 topics Subcollection 25

4.7 content_chunks Collection 26

4.8 student_progress Collection 26

4.9 student_schedules Collection 27

5.1 Test Case 1 47

5.2 Test Case 2 48

5.3 Test Case 3 49

5.4 Test Case 4 50

LIST OF FIGURES

4.1 Use Case Diagram 20

4.2 Student Learning Loop Activity Diagram 21

4.3 Class Diagram 22

4.4 Database Schema 23

4.5 Landing Screen 28

4.6 Login Screen 29

4.7 Student Registration 30

4.8 User Profile Management 31

4.9 Personalized Study Schedule 32

4.10 Subject List 33

4.11 Subject Details and Enrollment 34

4.12 Learning Content Viewer 35

4.13 AI-Generated Quiz 36

4.14 Student Schedule 37

4.15 Progress Tracking Dashboard 38

4.16 Knowledge Provider Dashboard 39

4.17 Syllabus and Study Material Upload 40

4.18 Admin Dashboard 41

4.19 Create New Subject 42

4.20 Topic and Content Management 43

4.21 User Management 44

ABBREVIATIONS

AI	Artificial Intelligence
ALS	Adaptive Learning System
API	Application Programming Interface
ITS	Intelligent Tutoring System
KP	Knowledge Provider
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL BACKGROUND

For decades, the standard educational model has followed a one-size-fits-all approach, where a single curriculum is delivered at a uniform pace to a diverse group of students. While effective for mass education, this model inherently struggles to cater to individual learning speeds, prior knowledge, and unique learning challenges. The discrepancy in learning outcomes was famously highlighted by Benjamin Bloom in his Two Sigma Problem, which demonstrated that students receiving one-on-one human tutoring performed two standard deviations better than those in a traditional classroom setting. The field of Educational Technology has long strived to bridge this Two Sigma gap by creating systems that can provide individualized attention at scale. Early attempts included Intelligent Tutoring Systems and Adaptive Learning Systems, which used rule-based algorithms and statistical models to guide students through predefined learning paths. While valuable, these systems were often expensive to develop, difficult to adapt to new subjects, and limited in their ability to engage in natural, unscripted dialogue with learners. The recent and rapid advancement of Large Language Models such as Google's Gemini presents a paradigm shift. This technological leap provides a novel opportunity to build a new generation of learning platforms that are not just adaptive, but truly intelligent, conversational, and deeply personalized. This project, is situated at the intersection of proven principles and cutting-edge AI technology, aiming to create a scalable solution that moves closer to the ideal of a personal human tutor for every student.

1.2 OBJECTIVE

The goal is to create a proof-of-concept that demonstrates the viability of using a Large Language Model to enhance student mastery of curriculum subjects by providing a dynamic learning path, AI-assisted tutoring, and relevant content, all tailored to individual performance and needs.

- **To Develop a Dynamic, Personalized Learning Path Generator:** The system will generate and continuously adapt a personalized study schedule for each student based on their initial knowledge, learning goals, weekly availability, and ongoing performance in assessments.
- **To Implement a Context-Aware AI Tutoring System:** The platform will feature an AI tutor, powered by the Gemini API, capable of providing real-time, context-aware explanations, answering student questions, and adapting its teaching style (e.g., simplifying concepts or providing more examples) based on detected student struggle.
- **To Create an AI-Assisted Content Ingestion and Management Workflow:** The project will implement a streamlined process for subject matter experts (Knowledge Providers) to upload educational materials (PDFs), which the AI will then automatically process into structured, digestible learning "chunks" for student consumption.
- **To Integrate On-the-Fly, AI-Generated Assessments:** The system will dynamically generate short quizzes based on the specific content a student has just learned, providing immediate, explanatory feedback to reinforce learning and identify knowledge gaps.
- **To Build a Functional Prototype with Distinct User Roles:** The final deliverable will be a working application that successfully integrates these AI features and supports the distinct workflows for the users.

CHAPTER 2

LITERATURE SURVEY

2.1 THE CHALLENGE OF PERSONALIZED LEARNING

The foundational motivation for adaptive educational technology is often traced to the "Two Sigma Problem." This seminal finding by Bloom [1] demonstrated that students receiving one-on-one human tutoring performed two standard deviations higher than those in traditional classroom instruction. Achieving this level of individualized support at scale has become the central challenge for the field. This survey examines the key technological paradigms developed to address this challenge and provides the context for current research in AI-driven educational technology.

2.2 PARADIGMS IN ADAPTIVE EDUCATIONAL SYSTEMS

The effort to replicate the benefits of one-on-one tutoring has led to two primary models: Adaptive Learning Systems (ALS) and Intelligent Tutoring Systems (ITS). An ALS, as defined by Brusilovsky [2], primarily focuses on dynamically adjusting the sequence of educational material based on a model of the student's knowledge. In contrast, an ITS aims to simulate a human tutor by engaging in interactive dialogue and providing explanatory feedback, a field historically challenged by the complexities of natural language understanding [3].

The advent of Large Language Models (LLMs) like Google's Gemini [5] repre-

sents a significant technological leap for both fields. However, a critical challenge with using general-purpose LLMs in an educational context is their propensity for hallucination providing factually incorrect or nonsensical information. To mitigate this risk, a prominent strategy is Retrieval-Augmented Generation (RAG). In this architecture, the LLM is provided with curated, trusted information retrieved from a specific knowledge base (e.g., a project database) before generating a response. This "grounding" technique constrains the model's output, ensuring its responses are factually accurate and aligned with a specific curriculum.

2.3 AI IN CONTENT AND ASSESSMENT GENERATION

A major bottleneck in creating digital learning experiences is the labor-intensive process of authoring high-quality content and assessments. Early attempts at Automatic Question Generation (AQG) were often limited by rule-based methods, resulting in pedagogically shallow outputs [4].

Modern LLMs have shown a remarkable capability for these tasks, able to generate varied question types, detailed feedback, and content summaries from source text. Consequently, a key area of application is the development of semi-automated workflows where LLMs assist subject matter experts. This often involves a human-in-the-loop system where an AI processes raw source materials (e.g., PDFs) into structured, digestible content chunks, which are then reviewed and validated by a human expert. Furthermore, these models enable the dynamic generation of formative assessments based on specific content a student has just reviewed, ensuring tight alignment between learning material and evaluation.

2.4 SYNTHESIS AND RESEARCH GAP

The literature shows that LLMs can individually enhance components of the educational process: they can power adaptive sequencing (like an ALS), facilitate conversational learning (like an ITS), and automate the generation of content and quizzes.

While significant progress has been made in these individual areas, a remaining research gap lies in the holistic integration of these components into a single, cohesive feedback loop. Such a system would leverage a unified AI engine to orchestrate an end-to-end learning process where:

1. AI assists in curating a verified knowledge base from expert-provided materials.
2. This trusted knowledge base is used to ground both a conversational tutor and a dynamic assessment generator, ensuring accuracy and relevance.
3. Performance data from the assessments directly informs an adaptive scheduling engine.
4. The scheduling engine, in turn, modifies the student's learning path in real-time.

The development of such a tightly-integrated, end-to-end adaptive system represents a key challenge and a significant opportunity in the evolution of personalized learning technology.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing system for a typical student is not a single piece of software but a composite of traditional methods and disparate digital tools. This fragmented ecosystem presents several significant drawbacks that hinder effective and personalized learning.

Traditional Methods: Classroom lectures, physical textbooks, and study groups form the baseline. While valuable, they are inherently designed for a general audience and operate at a fixed pace, making it difficult to cater to individual learning speeds.

Static Digital Repositories: Learning Management Systems (LMS) like Moodle, Canvas, or Google Classroom are widely used. However, their primary function is content hosting and organization. They act as digital file cabinets where educators can upload PDFs, presentations, and assignments. They lack intelligence, do not adapt to student performance, and offer a passive learning experience.

One-Size-Fits-All Online Content: Platforms like YouTube and Khan Academy provide a wealth of high-quality educational videos and articles. While excellent as supplementary resources, they do not offer a personalized learning path. Students are left to navigate vast libraries of content on their own, with no system to guide them based on their specific needs or knowledge gaps.

First-Generation Adaptive Systems: Existing adaptive learning platforms have made strides in personalization but often rely on rigid, pre-authored decision trees and proprietary content. They may lack the flexibility to incorporate an instructor's specific

materials or to engage in the kind of natural, conversational tutoring that is critical for deeper understanding.

The key limitations of this existing ecosystem can be summarized as follows:

- **Lack of Deep Personalization:** The learning path is static and not tailored to the individual's real-time performance, prior knowledge, or personal schedule.
- **Passive Learning Experience:** Students are primarily consumers of information (reading PDFs, watching videos). The feedback loop is slow, with assessments often being summative rather than formative and immediate.
- **Inflexible and Unrealistic Scheduling:** Students are solely responsible for managing their study time. No system assists in creating a realistic schedule that adapts when life events occur or when a topic proves more difficult than anticipated.
- **High Cost and Lack of Scalability for True Tutoring:** The gold standard, one-on-one human tutoring, is prohibitively expensive and not scalable to all students.

3.2 PROPOSED SYSTEM

The proposed system, AuraLearn, is designed to directly address the limitations of the existing systems by creating a holistic, intelligent, and deeply personalized learning environment. It leverages a Large Language Model (Google Gemini) as its core engine to deliver a dynamic and interactive experience.

The proposed system is built on four key functional pillars:

- **Personalized Adaptive Scheduling:** AuraLearn moves beyond static syllabi. Upon enrollment, a student provides their initial knowledge level, learning goals, and weekly availability. The AI uses this information to generate a realistic, personalized study roadmap. Critically, this schedule is not fixed. As the student

completes quizzes, the system's adaptive engine analyzes their performance and dynamically adjusts the schedule, either allocating more time for difficult topics or accelerating through mastered content. This directly solves the problem of inflexible and unrealistic scheduling.

- **Interactive AI Tutoring and Learning:** To combat passive learning, AuraLearn provides an active learning environment. Students study from high-quality, AI-processed content_chunks that are approved by subject matter experts. At any point, they can engage with a conversational AI tutor. This tutor is grounded in the platform's verified content (using a RAG model) to ensure accuracy. It can answer questions, provide simpler explanations, offer new examples, and guide students through their difficulties, simulating the immediate and responsive nature of a human tutor.

- **AI-Powered Content Ingestion:** AuraLearn simplifies the process of creating structured learning content. Knowledge Providers (KPs) can upload their existing educational materials (e.g., PDF textbooks or notes). The system's AI then automatically processes this material, segmenting it by topic and breaking it down into small, digestible content_chunks. This workflow empowers educators to easily bring their own high-quality content into an adaptive environment, solving the issue of being locked into proprietary content.

- **Formative Assessment:** AuraLearn integrates assessment directly into the learning process. After studying a content_chunk, a student can immediately take a short quiz that is generated by the AI, based on the material they just reviewed. The system provides instant feedback, including explanations for why an answer was correct or incorrect. This creates a tight, continuous feedback loop that reinforces learning and provides the performance data needed for the adaptive scheduling engine.

3.3 MODULE DESCRIPTION

The AuraLearn system is architected as a set of distinct, interconnected modules, each responsible for a specific domain of functionality. This modular design ensures clarity in development and allows for a scalable and maintainable system.

1. **Curriculum and Knowledge Graph Module**

   This module serves as the foundational architectural layer of the platform, defining the academic blueprint for all learning activities. It is primarily managed by the Administrator. Its key functions include: Subject Management, Syllabus Ingestion, Topic Organization, and the Knowledge Graph Editor.

2. **Content Life Cycle Module**

   This module manages the end-to-end workflow for transforming raw educational materials into structured, validated learning content. It is a collaborative module used by both Knowledge Providers and Administrators. Its functions include: Material Upload, AI Chunking, and Content Review and Approval.

3. **User and Access Module**

   This module is responsible for the critical functions of security, authentication, and user identity management. Its functions include: Authentication, User Profile Management, and Role-Based Access Control.

4. **Learning and Assessment Module**

   This module constitutes the primary interactive interface for the student. Its functions include: Content Viewer, AI Tutor Chat, AI Quiz Generator, and a Feedback System.

5. **Adaptive Scheduling Module**

   This module is the personalization engine of AuraLearn. Its functions include: an Initial Schedule Generator, Dynamic Scheduling logic, and a Progress Tracker.

6. **Admin Control Module**

   This module provides the centralized command center for an Administrator. Its functions include: User Management, Content Quality Control, and Curriculum and Content Assignment.

3.4 FEASIBILITY STUDY

A feasibility study is conducted to assess the viability of the AuraLearn project across operational, technical, and economic dimensions.

3.4.1 Operational Feasibility

Operational feasibility assesses how well the system will be accepted and used by its intended actors. AuraLearn is designed to directly improve the workflows for all user roles.

- **For Students:** The system offers a flexible, personalized learning path that adapts to their schedule and performance, directly addressing the limitations of traditional, one-pace-fits-all education.
- **For Knowledge Providers:** The AI-assisted content ingestion pipeline automates the time-consuming task of structuring learning materials, allowing experts to focus on quality control and content validation.
- **For Administrators:** Centralized dashboards provide clear oversight of users, subjects, and content quality, simplifying platform management.

Conclusion: The system is operationally feasible. By providing clear value and a user-friendly experience for each role, the platform is expected to achieve high user acceptance.

3.4.2 Technical Feasibility

Technical feasibility assesses if the project can be successfully developed using current technology and expertise.

- **Technology Stack:** The chosen stackFlutter for the cross-platform frontend, and Firebase (Firestore, Authentication) with a Node.js proxy for the backendis modern, well-documented, and highly suitable for building a scalable, real-time application.
- **Core Challenge:** The primary technical challenge lies in prompt engineering and the effective implementation of a Retrieval-Augmented Generation (RAG) model with the Google Gemini API. This is a recognized part of the development process, not an insurmountable barrier.
- **Risk Mitigation:** Key risks include AI hallucinations, which are mitigated by grounding the AI in a human-approved knowledge base, and database security, which will be managed via granular Firebase Security Rules.

Conclusion: The project is technically feasible. The required technologies are readily available, and the core technical challenges are well-defined with clear mitigation strategies.

3.4.3 Economic Feasibility

Economic feasibility evaluates the project's financial viability as a commercial product.

- **Cost Structure:** Primary operational costs are driven by scalable cloud infrastructure (Firebase) and direct AI API usage (Google Gemini), which increases with user engagement, alongside ongoing personnel expenses.

- **Revenue Model:** Revenue can be generated through recurring models, such as B2C subscriptions for individual students and B2B institutional licensing for educational institutions.

Conclusion: The project is economically feasible. Its high-value personalized learning service supports strong recurring revenue models, providing a clear path to cover scalable operational costs and achieve profitability in a competitive market.

3.5 SYSTEM ENVIRONMENT

The system environment defines the hardware, software, and network components required for the development and operation of the AuraLearn platform.

Hardware:

- *Development Machine:* A standard desktop or laptop computer.
- *User Device:* Modern Android/iOS smartphones, tablets, or web browsers.

Software:

- *IDE:* Visual Studio Code or Android Studio.
- *Frameworks:* Flutter 3.35.3
- *Backend Services:* Firebase (Firestore, Authentication).
- *External APIs:* Google Gemini API.
- *Version Control:* Git.

Network: A stable internet connection is required for both developers (to access services) and end-users (to use the application).

3.6 ACTORS AND THEIR ROLES

An actor is an entity that interacts with the AuraLearn platform. There are four key actors in the system.

1. **Student**

   The primary end-user. Their goal is to master a subject through a personalized and interactive learning journey. Key responsibilities include following the schedule, consuming content, interacting with the AI tutor, and taking quizzes.

2. **Knowledge Provider (KP)**

   A subject matter expert. Their goal is to ensure the platform's knowledge base is accurate. Key responsibilities include uploading source materials and reviewing/approving AI-processed content.

3. **Administrator (Admin)**

   A power user with oversight of the entire platform. Their goal is to manage the system's structure, user base, and content quality. Key responsibilities include managing subjects, topics, users, and performing final content approval.

CHAPTER 4

SYSTEM DESIGN

4.1 METHODOLOGY

The development of AuraLearn will follow the Agile Scrum framework, selected for its suitability for projects with evolving requirements, such as the integration of AI features. This approach emphasizes flexibility and delivering working software in small, iterative cycles (sprints), which is ideal for AuraLearn as it allows for experimentation and adaptation of its AI-driven functionalities.

The core components of the Scrum framework utilized are:

Product Owner Defines the project vision and prioritizes features in the Product Backlog, ensuring alignment with user and business needs.

Development Team Builds and tests the software increments during each sprint, self-organizing to achieve sprint goals.

Scrum Master Facilitates the process, coaches the team on Scrum principles, and removes any impediments to progress.

Product Backlog A comprehensive, prioritized list of all features, requirements, enhancements, and bug fixes for the project.

Increment The shippable, potentially releasable version of the software produced at the end of each sprint, representing the sum of all completed Product Backlog items from current and previous sprints.

4.2 USER STORY

ID	As a (Type of Users)	I want to (Perform some task)	So that I can (Achieve some goal)
1	a user	register for an account and login securely	access the platform features relevant to my role.
2	an Admin	manage all user accounts and their roles	maintain the security and organization of the platform.
3	an Admin	create, manage, and assign subjects to Knowledge Providers	establish the platform's course structure and clearly delegate content responsibility.
4	a Knowledge Provider	upload a subject syllabus	have the system automatically generate the topic list.
5	a Knowledge Provider	review and manage AI-generated topics	ensure the curriculum is accurate.
6	a student	view and manage my basic profile	keep my account details up to date.
7	a Knowledge Provider	upload study material for a specific subject	have the AI process it into learning content.
8	a student	input my learning goals and weekly availability	have the platform generate a personalized study schedule tailored to me.
9	a student	view clear focused learning contents for my current topic	understand the material without distractions.

ID	As a (Type of Users)	I want to (Perform some task)	So that I can (Achieve some goal)
10	a student	ask questions in a chat interface and receive AI-powered help	resolve confusion without waiting for a human tutor.
11	a student	take a short quiz after studying	test my understanding and validate my knowledge.
12	a student	get instant feedback on my quiz answer with explanations	learn from my mistakes immediately.
13	a student	have my schedule automatically update based on quiz performance	focus more on topics that are difficult and speed through ones I master.
14	a student	see a visual dashboard of my progress	stay motivated and track my achievements.

Table 4.1: AuraLearn User Stories

4.3 PRODUCT BACKLOG

User Story ID	Priority	Size	Sprint	Status	Release Date	Release Goal
1	High	3	1	Completed	12/08/25	Enable secure user registration and login.
2	Medium	3		Completed	12/08/25	Allow Admins to manage user accounts and roles.
3	Medium	2		Completed	19/08/25	Enable Admins to define subjects and assign content uploader.
4	High	5		Completed	19/08/25	Automate topic creation by processing syllabus.
5	Medium	2	2	Completed	26/08/25	Ensure curriculum accuracy by allowing Knowledge Provider to review AI content.
6	Low	1		Completed	26/08/25	Allow users to keep their account details up to date.
7	High	5		Completed	08/09/25	Ingest raw study material for the AI to process.
8	High	4	3	Completed	16/09/25	Create a personalized schedule based on student goals.

User Story ID	Priority	Size	Sprint	Status	Release Date	Release Goal
9	High	2		Completed	16/09/25	Provide a clean interface for students to study content.
10	High	5		Completed	23/09/25	Provide immediate on-demand help via an AI Tutor.
11	High	4		Completed	23/09/25	Allow students to test understanding with on-the-fly quizzes.
12	Medium	3	4	Completed	30/09/25	Help students learn from mistakes with instant quiz feedback.
13	High	4		Completed	30/09/25	Create a truly adaptive experience by updating the schedule.
14	Medium	3		Completed	07/10/25	Motivate students by visualizing their progress and achievements.

Table 4.2: AuraLearn Product Backlog

4.4 PROJECT PLAN

ID	Task Name	Start Date	End Date	Days
1	Implement User Registration & Login	11/08/25	12/08/25	2
2	Build Admin - User and Role Management UI	11/08/25	12/08/25	2
3	Build Admin - Subject and KP Assignment UI	13/08/25	19/08/25	3
4	Develop KP Syllabus Upload and AI Topic Generation	14/08/25	19/08/25	3
5	Build AI generated Topic Review UI	25/08/25	26/08/25	2
6	Implement Student Profile Management	25/08/25	26/08/25	2
7	Develop KP Study Material Upload & AI Chunking Pipeline	29/09/25	08/09/25	5
8	Implement Student Onboarding and Initial AI Schedule	11/09/25	16/09/25	4
9	Implement Student Content Viewer	15/09/25	16/09/25	2
10	Implement Student AI Tutor Chat Interface	17/09/25	23/09/25	5
11	Implement AI Quiz Generation	19/09/25	23/09/25	3
12	Implement AI Quiz Feedback for Student	26/09/25	30/09/25	3
13	Implement Adaptive Scheduling Logic for Student	26/09/25	30/09/25	3
14	Implement Student Progress Tracking Dashboard	06/10/25	07/10/25	2

Table 4.3: AuraLearn Project Plan

4.5 UML DIAGRAMS

4.5.1 Use Case Diagram

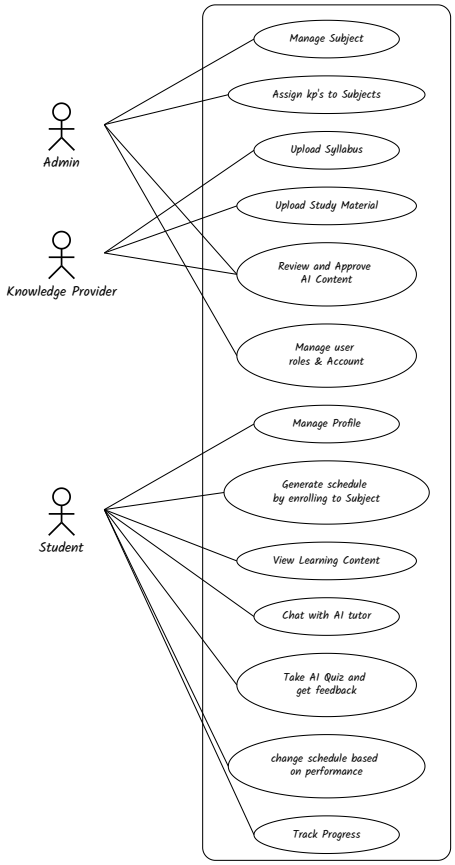

Figure 4.1: Use Case Diagram

4.5.2 Activity Diagram

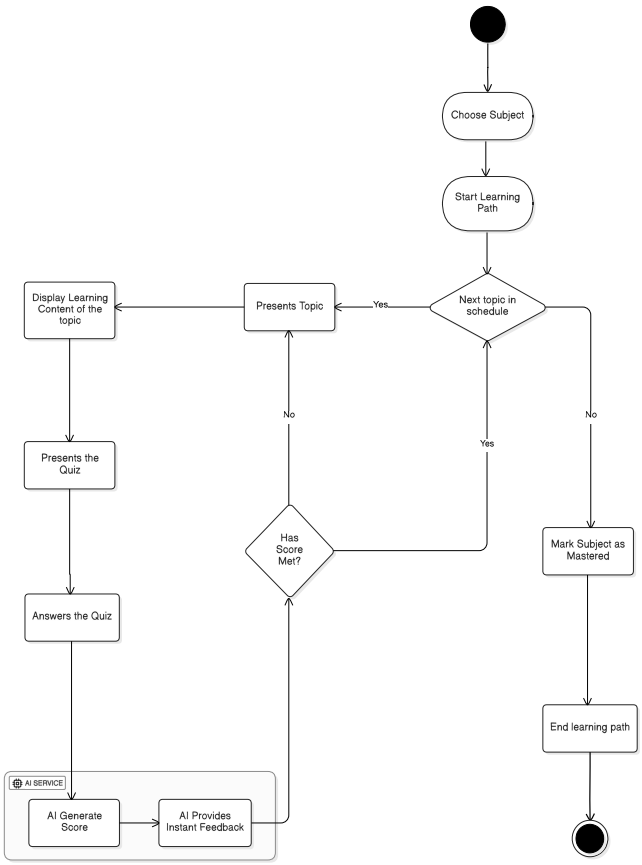

Figure 4.2: Student Learning Loop Activity Diagram

4.5.3 Class Diagram

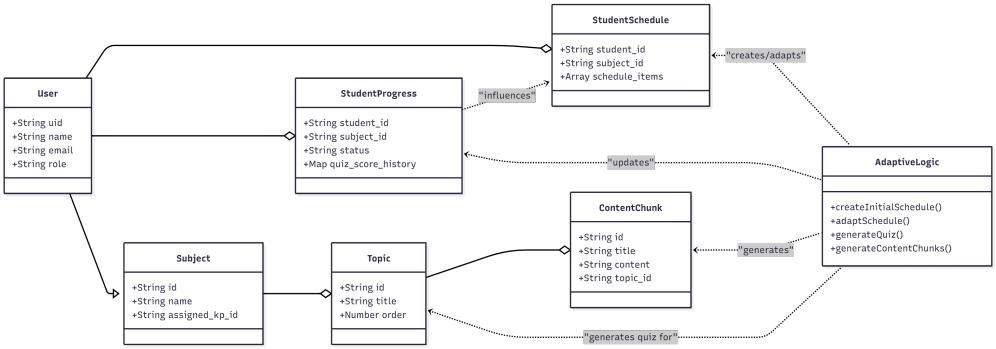

Figure 4.3: Class Diagram

4.6 DATABASE DESIGN

The backend of AuraLearn is built upon Google Firebase, utilizing its Cloud Firestore NoSQL database for flexible, scalable data storage and its dedicated Authentication service for secure user identity management. This design supports the dynamic and role-based nature of the application.

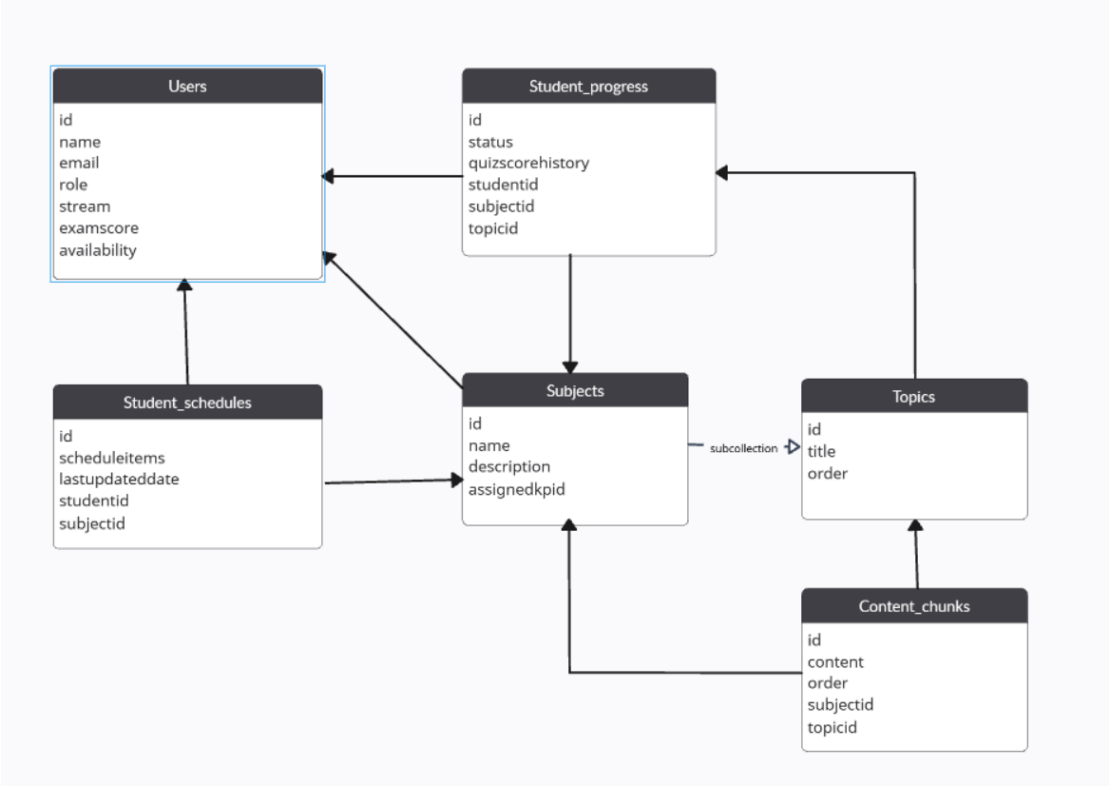

Figure 4.4: Database Schema

users Collection

Stores user profile data, linked to an account by its UID.

Field Name	Data Type	Description
id	string	Unique user ID from Firebase Authentication.
name	string	User's full name.
email	string	User's unique email address.
role	string	User role (Student, KP, Admin, or SuperAdmin).
createdAt	timestamp	The date and time the user account was created.
stream	string	(Student) Academic stream or major (e.g., 'BSc', 'MCA').
examScore	number	(Student) The score from their last relevant exam.
availability	map	(Student) Preferred weekly study days and time range.

Table 4.4: users Collection

subjects Collection

Defines the academic subjects offered on the platform and tracks their content lifecycle.

Field Name	Data Type	Description
id	string	The unique identifier for the subject (Document ID).
name	string	Official name of the subject.
description	string	Brief summary of the subject.

assignedKpId	string	UID of the Knowledge Provider assigned to this subject.

Table 4.5: subjects Collection

topics (Subcollection)

Represents individual learning topics within a subject.

Field Name	**Data Type**	**Description**
id	string	The unique identifier for the topic (Document ID).
title	string	The title of the topic.
description	string	A brief, AI-generated description of the topic.
order	number	The sequence number defining the learning order.

Table 4.6: topics Subcollection

content_chunks Collection

Stores the AI-generated, digestible units of learning content for each topic.

Field Name	**Data Type**	**Description**
id	string	The unique identifier for the content chunk (Document ID).
title	string	AI-generated title for the chunk.
content	string	The main text content of the learning chunk.
order	number	The reading order of chunks within a topic.

topicId	string	The ID of the parent topic this chunk belongs to.
subjectId	string	The ID of the parent subject.

Table 4.7: content_chunks Collection

student_progress Collection

Tracks each student's mastery and interaction with topics.

Field Name	Data Type	Description
id	string	The unique identifier for this progress record (Document ID).
studentId	string	The student's unique ID.
subjectId	string	The ID of the relevant subject.
topicId	string	The ID of the topic being tracked.
status	string	Mastery status (not_started, in_progress, mastered).
quizScoreHistory	map	A map of quiz scores with timestamps as keys.

Table 4.8: student_progress Collection

student_schedules Collection

Stores the personalized learning plan for each student, organized by subject.

Field Name	Data Type	Description
id	string	The unique identifier for this schedule document (Document ID).
studentId	string	The student's unique ID.

subjectId	string	The ID of the subject this schedule applies to.
scheduleItems	array of maps	An array of planned study sessions, each containing a date, topic ID, and topic name.
lastUpdatedDate	timestamp	The timestamp of the last AI-driven schedule update.

Table 4.9: student_schedules Collection

4.7 USER INTERFACE DESIGN

Landing Screen

The entry point of the application, featuring key information and navigation options for new and returning users.

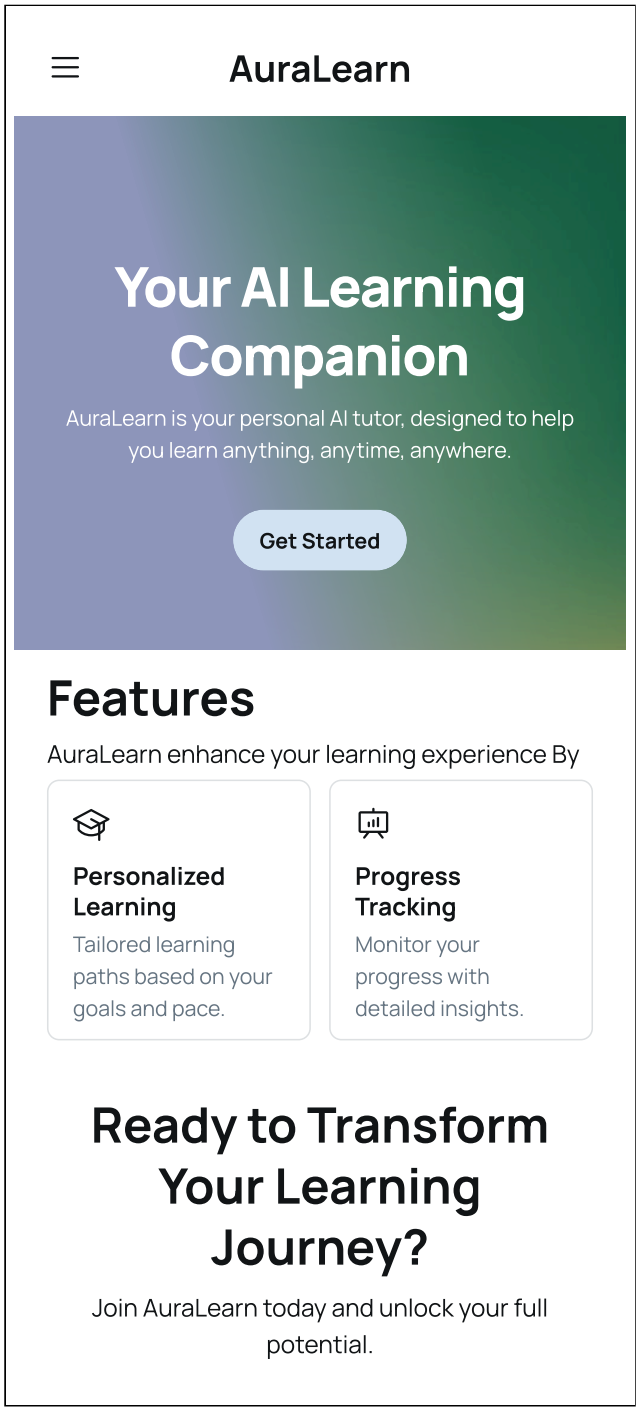

Figure 4.5: Landing Screen

Login Screen

A secure interface for users to access their accounts with credentials. It also includes links for new registrations and password recovery.

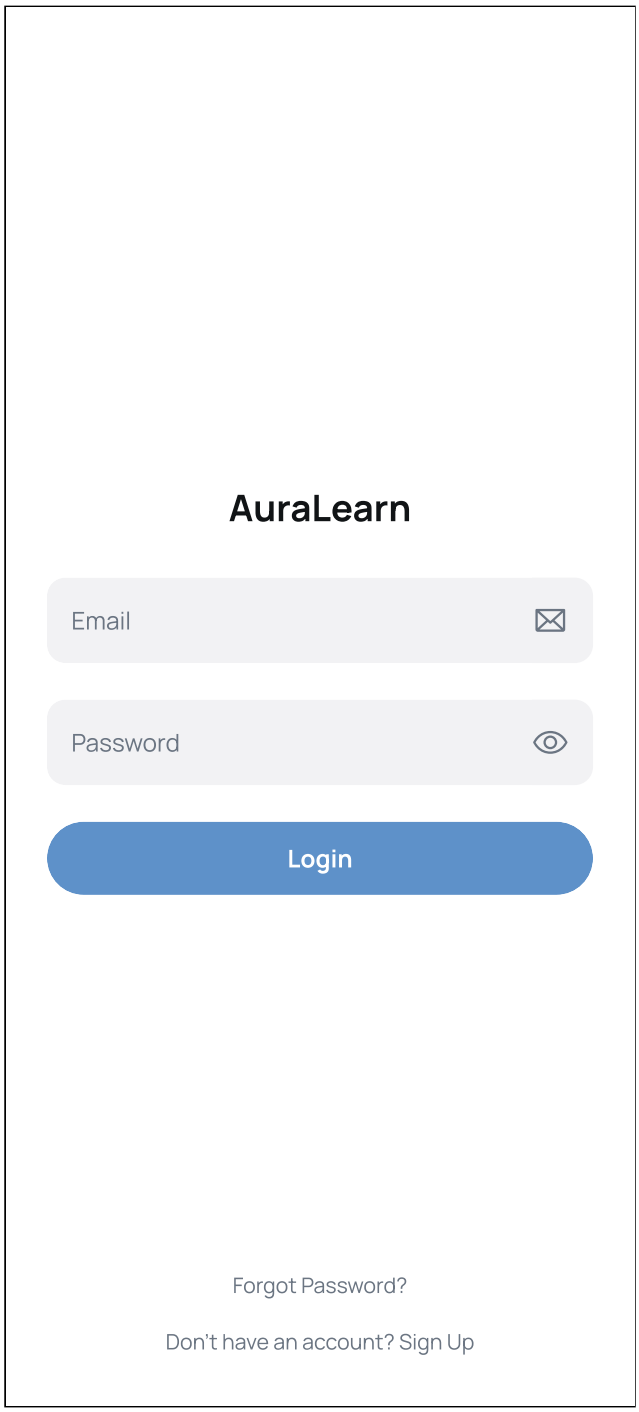

Figure 4.6: Login Screen

Student Registration

A form for new students to create an account, collecting necessary information to personalize their learning experience.

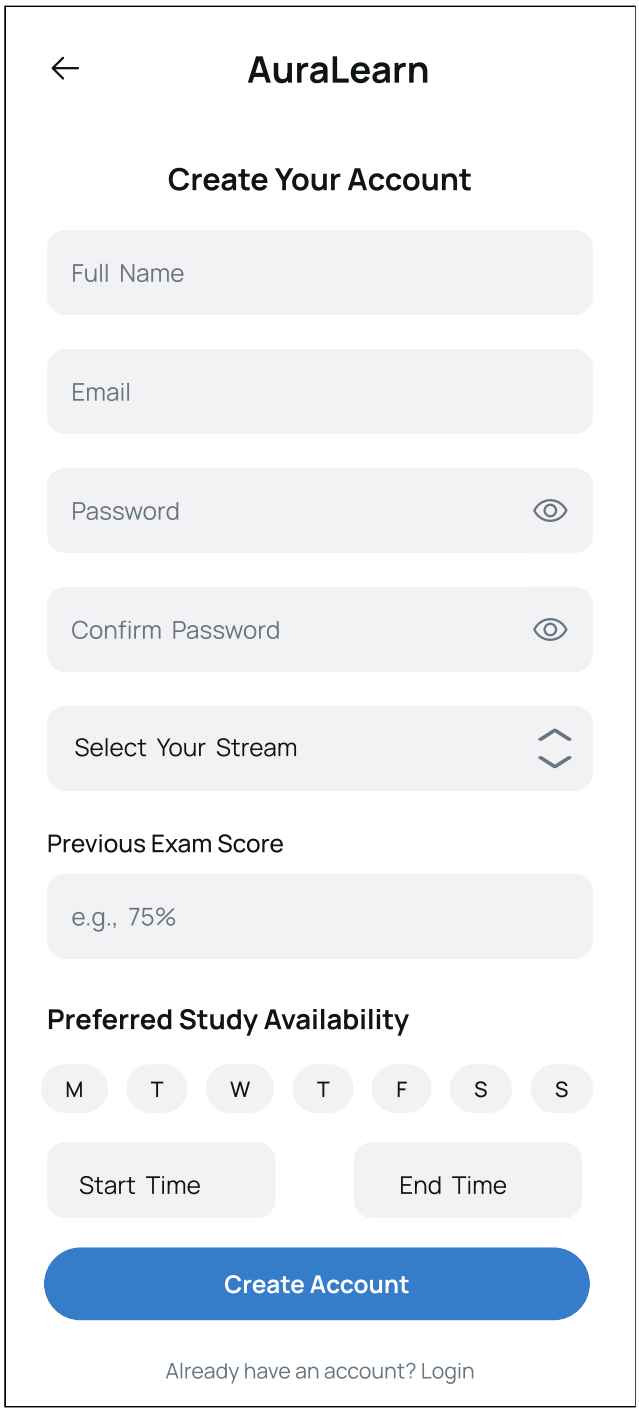

Figure 4.7: Student Registration

User Profile Management

An interface where users can view and update their personal details, manage their settings, and track account information.

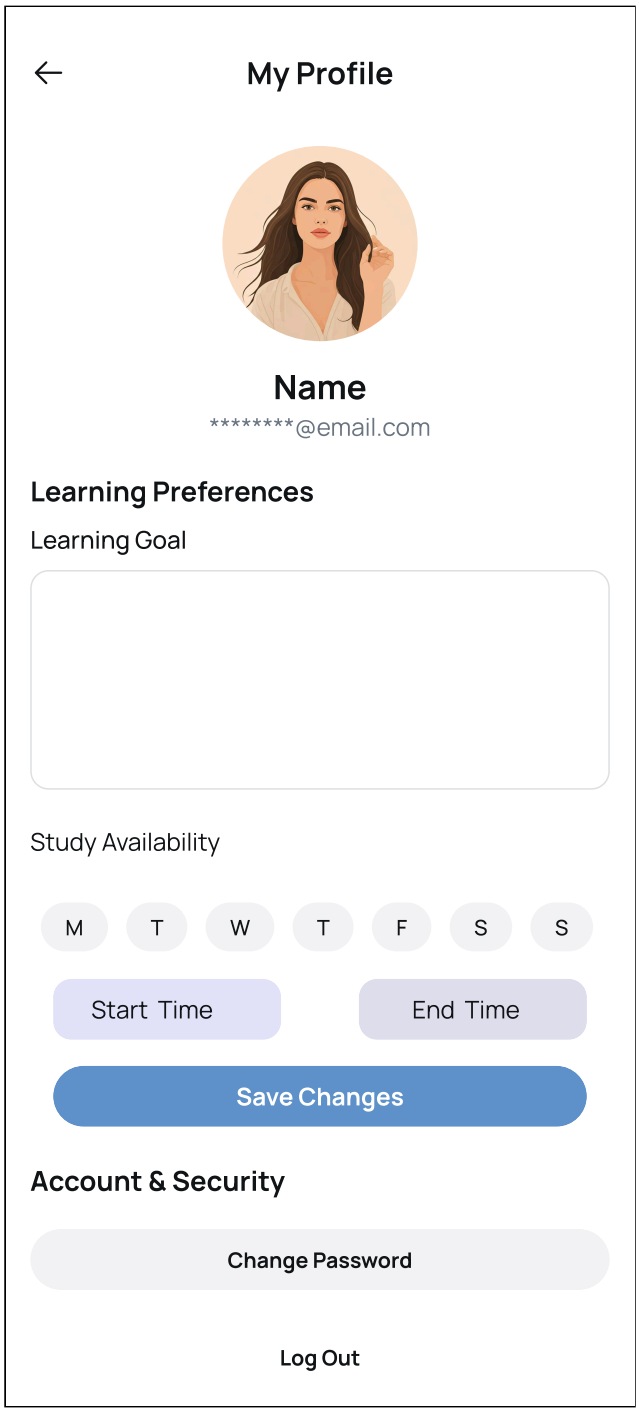

Figure 4.8: User Profile Management

Personalized Study Schedule

The dashboard for students, showing a personalized schedule and upcoming topics.

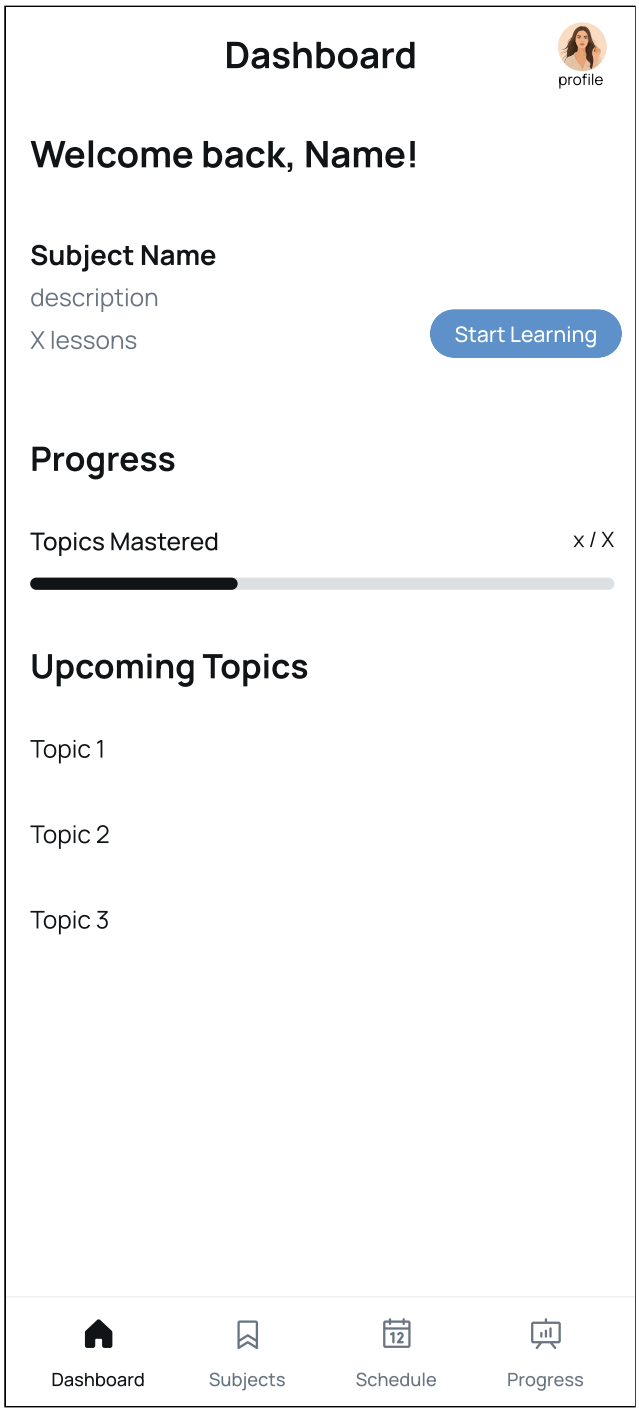

Figure 4.9: Personalized Study Schedule

Subject List

A comprehensive list of all available subjects, allowing students to browse and enroll in courses.

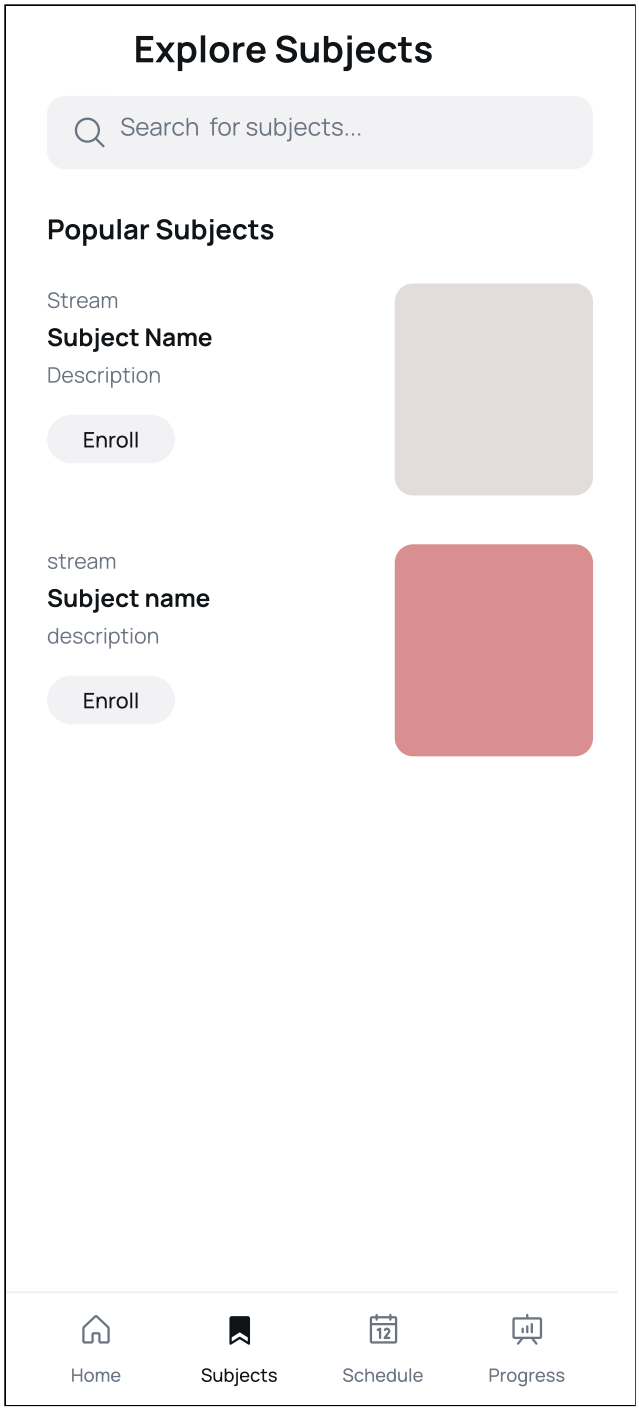

Figure 4.10: Subject List

Subject Details and Enrollment

Detailed information about a specific subject, including syllabus, topics, and an enrollment option.

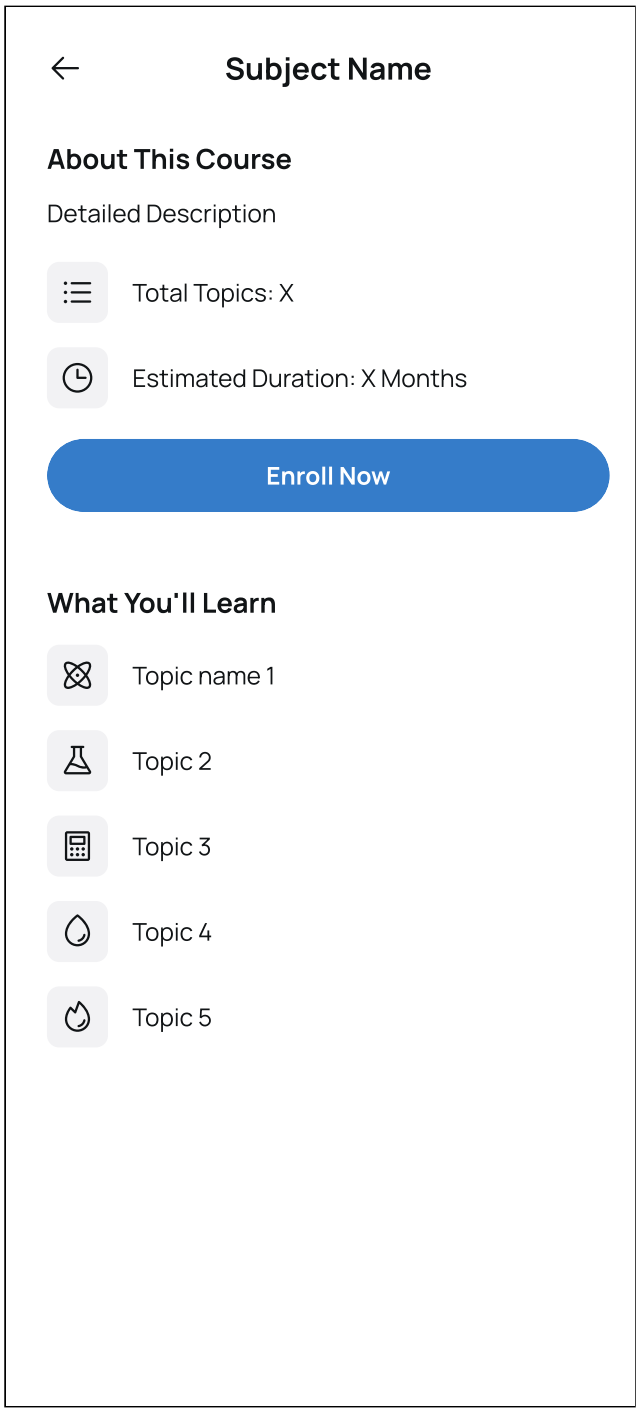

Figure 4.11: Subject Details and Enrollment

Learning Content Viewer

The interface for viewing learning materials with navigation and interactive features.

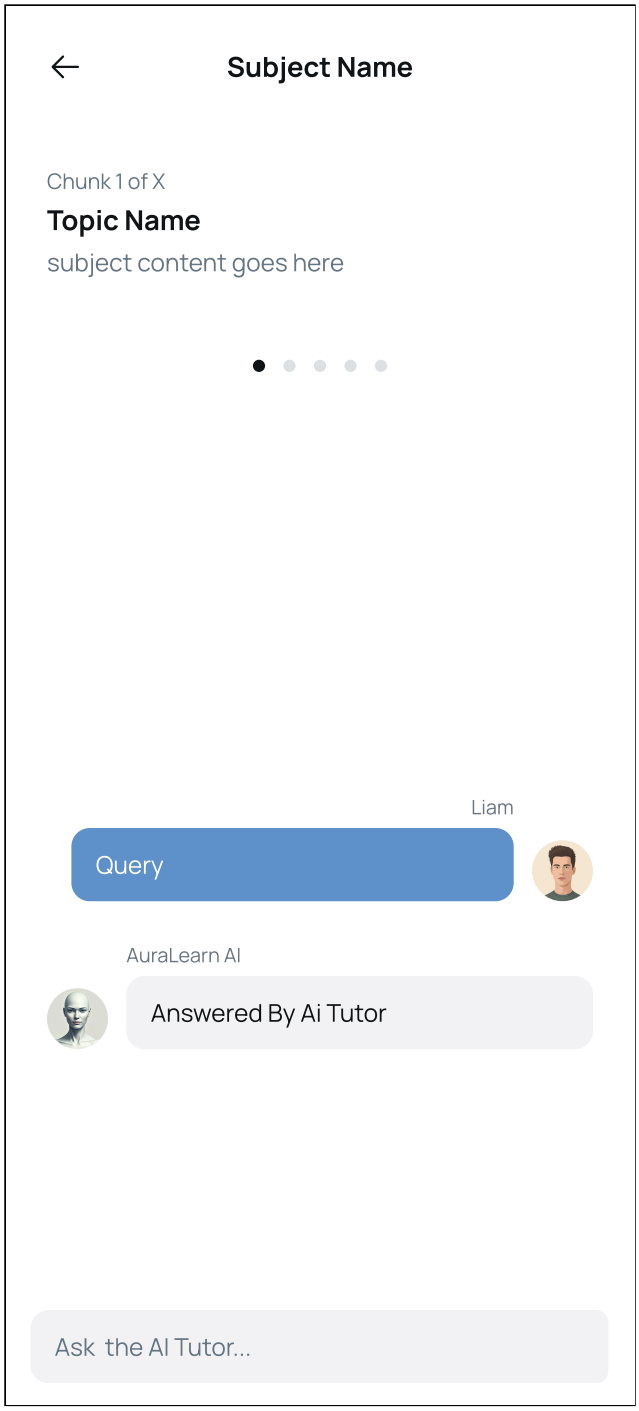

Figure 4.12: Learning Content Viewer

AI-Generated Quiz

An interactive quiz interface that presents questions generated by an AI with immediate feedback.

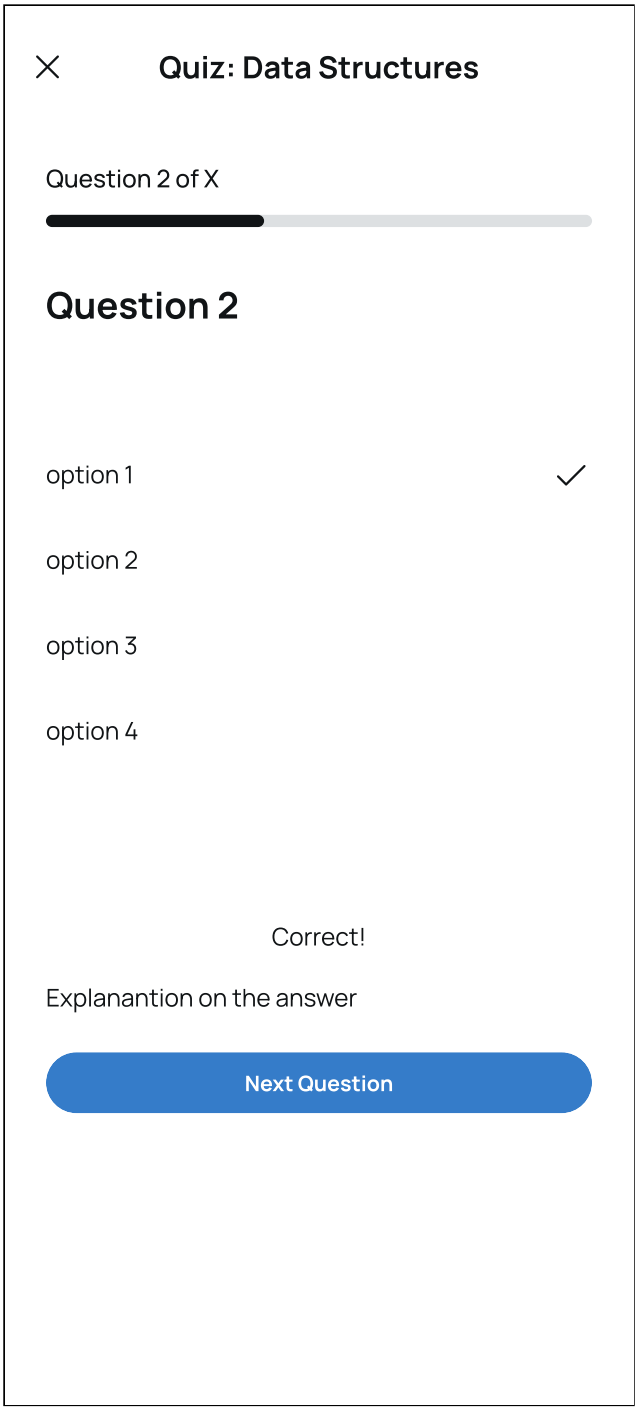

Figure 4.13: AI-Generated Quiz

Student Schedule

An interface for students to view their personalized study schedule.

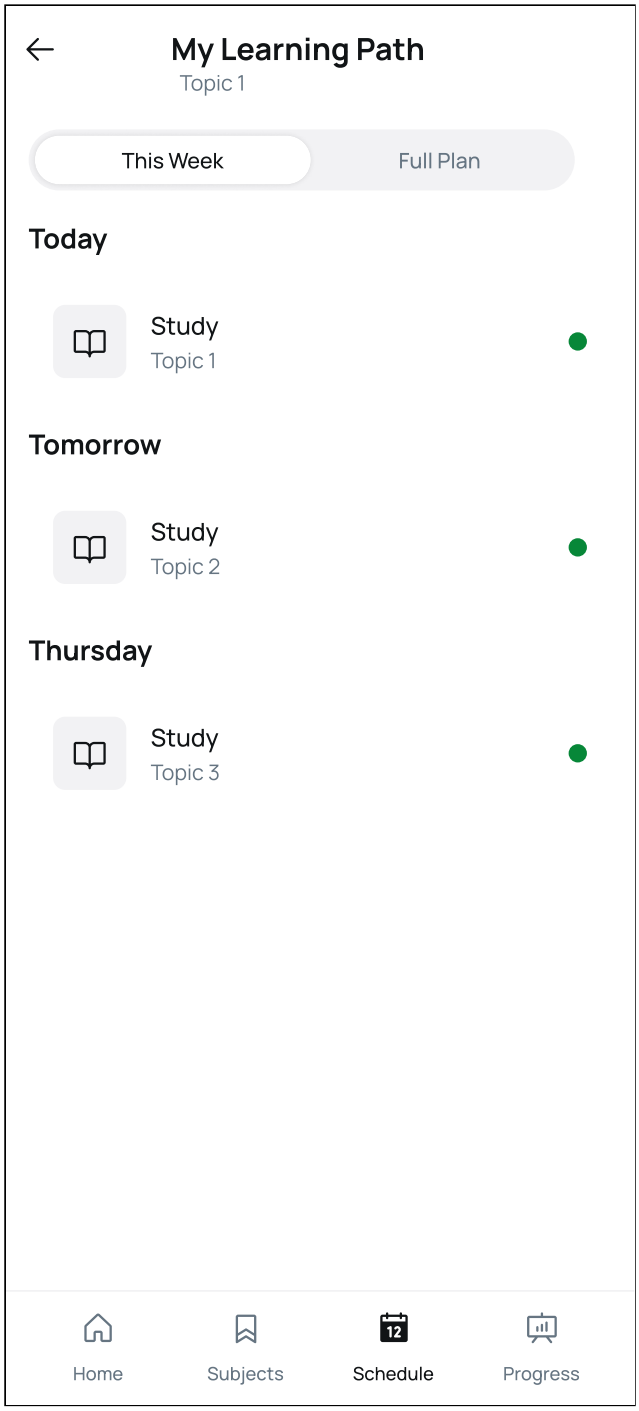

Figure 4.14: Student Schedule

Progress Tracking Dashboard

A visual representation of a student's learning progress, including completed topics and scores.

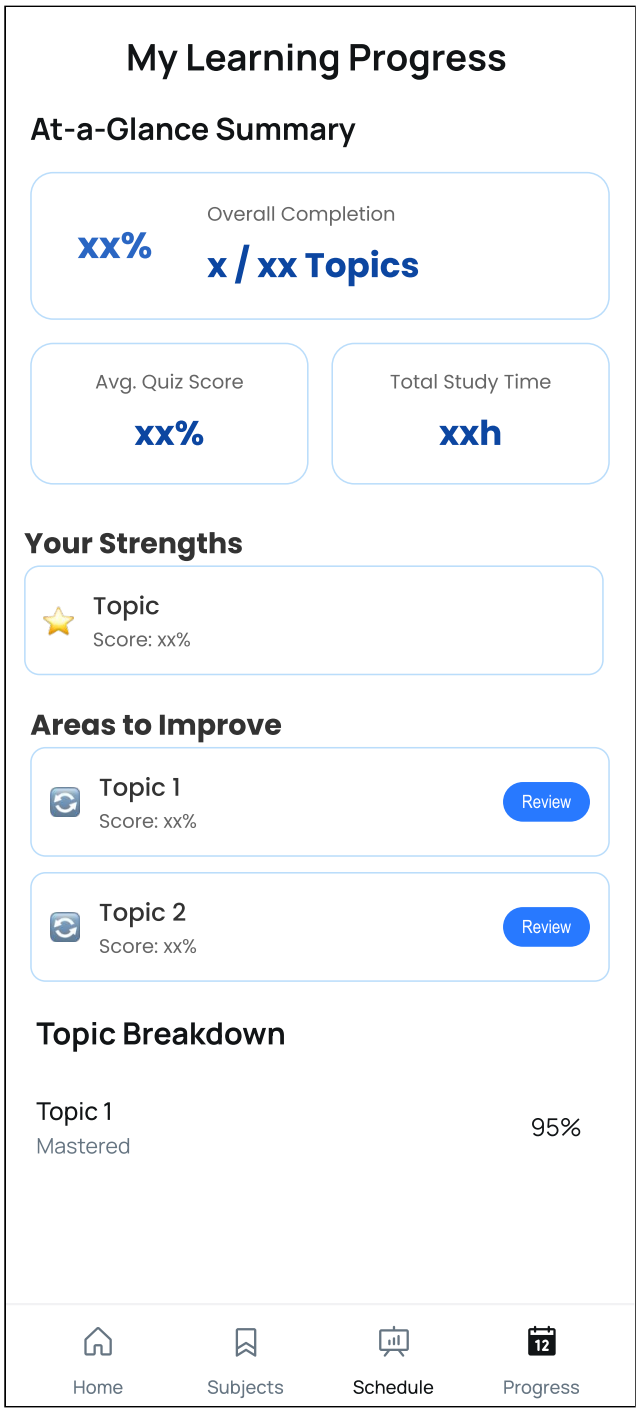

Figure 4.15: Progress Tracking Dashboard

Knowledge Provider Dashboard

The main dashboard for Knowledge Providers to manage their subjects and learning materials.

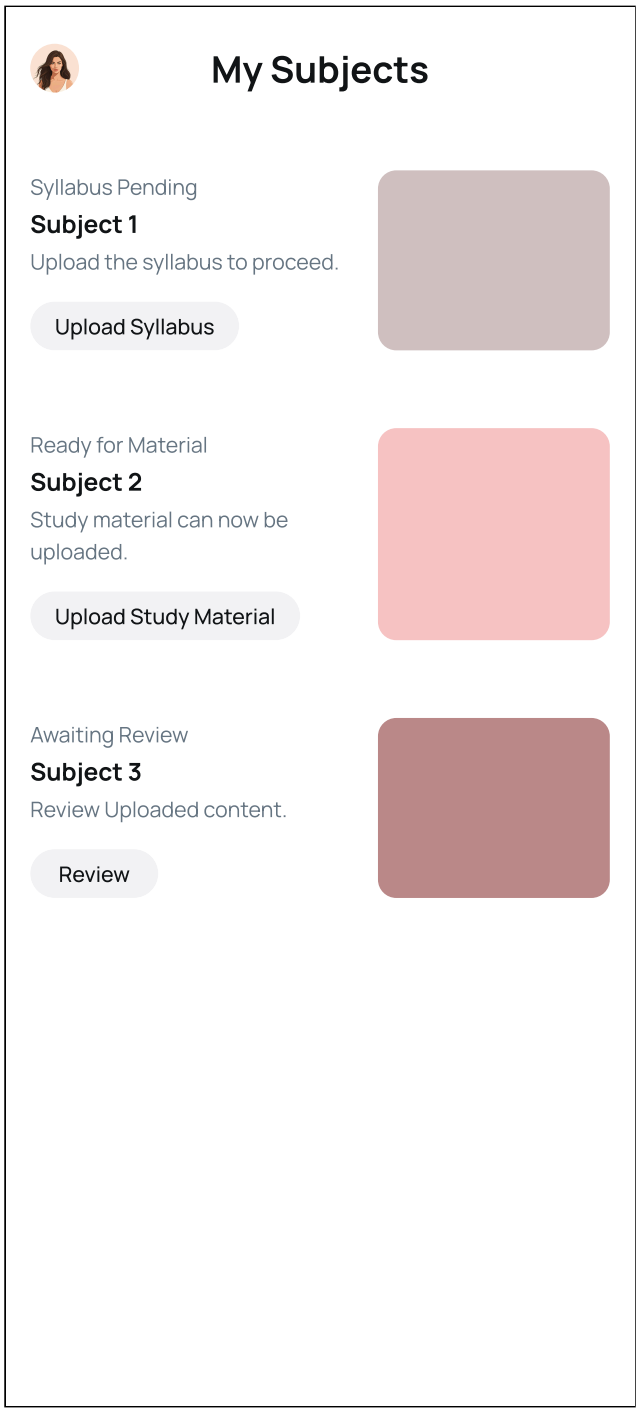

Figure 4.16: Knowledge Provider Dashboard

Syllabus and Study Material Upload

An interface for Knowledge Providers to upload syllabus and study materials for a subject.

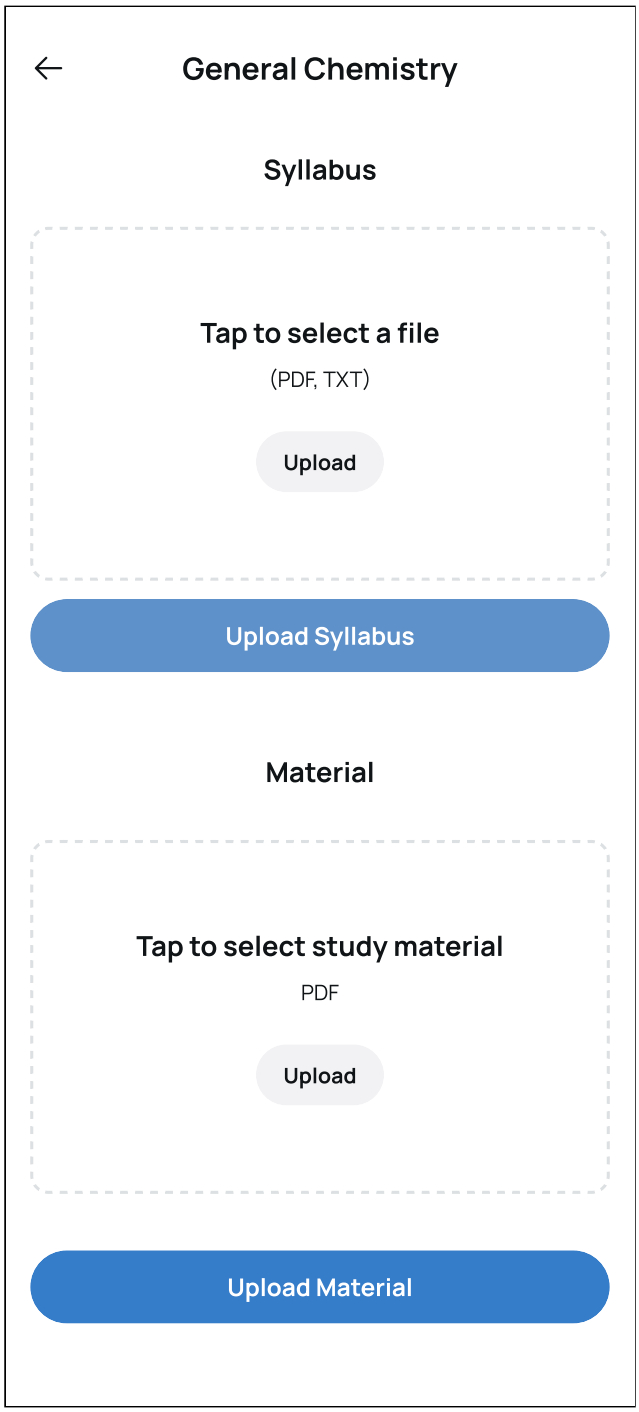

Figure 4.17: Syllabus and Study Material Upload

Admin Dashboard

The main control panel for administrators, offering an overview of user activity and management tools.

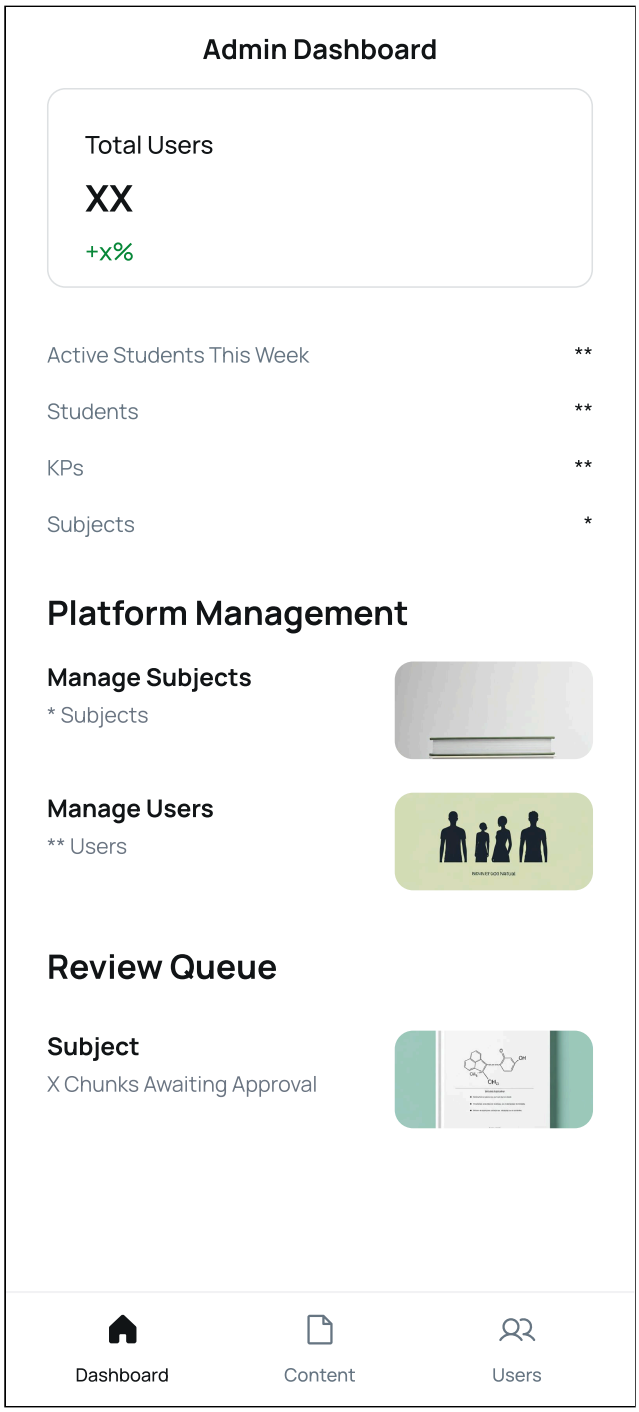

Figure 4.18: Admin Dashboard

Create New Subject

An administrative interface to add new subjects and assign them to Knowledge Providers.

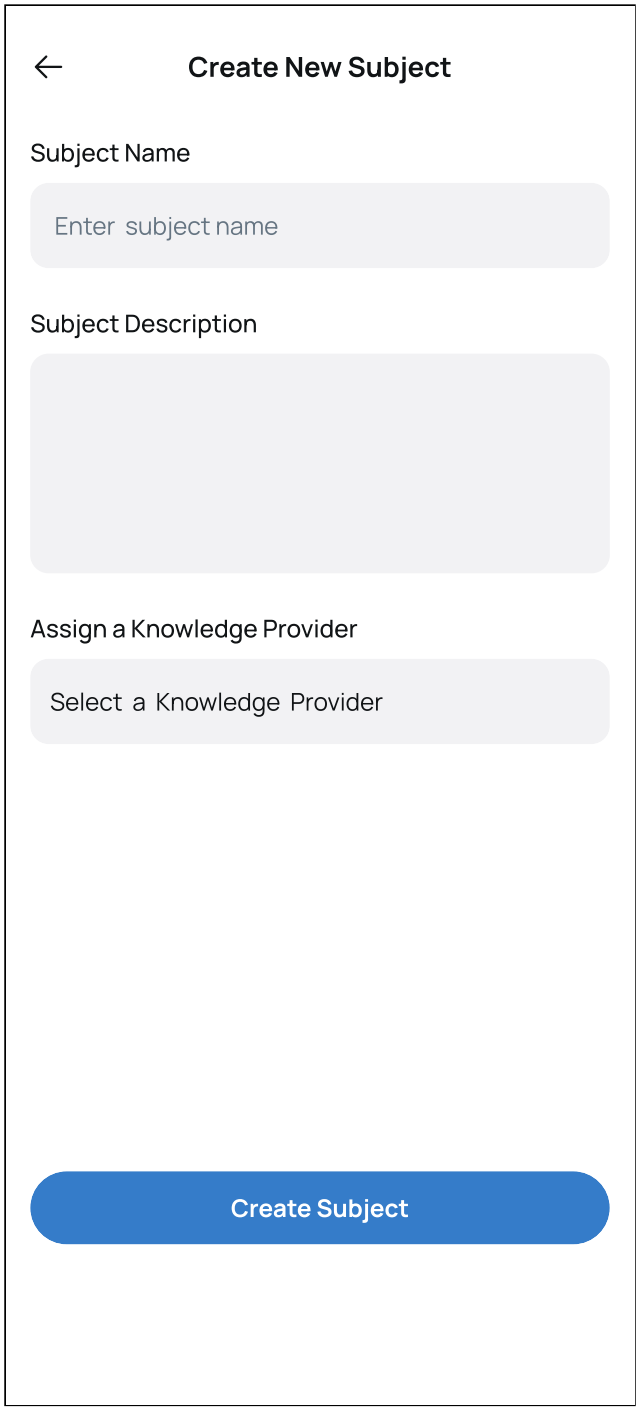

Figure 4.19: Create New Subject

Topic and Content Management

A screen for admins to organize, edit, and update the topics and content for each subject.

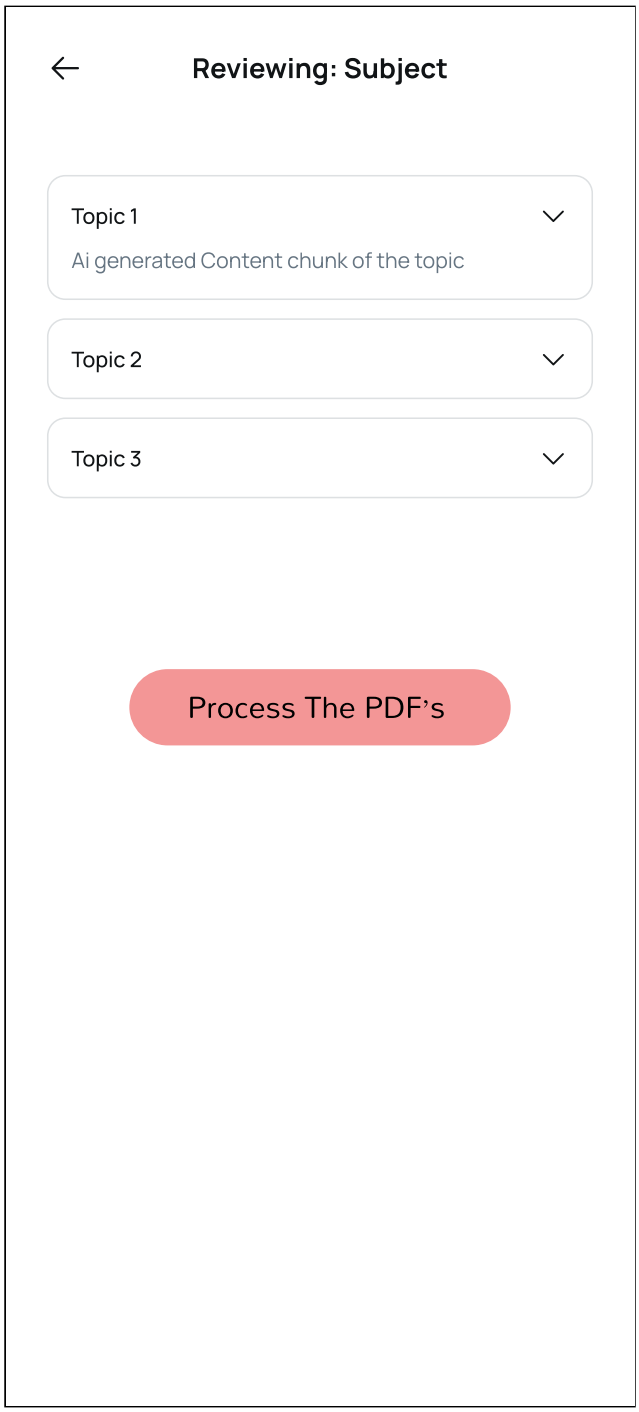

Figure 4.20: Topic and Content Management

User Management

A tool for administrators to manage user accounts, roles, and permissions across the platform.

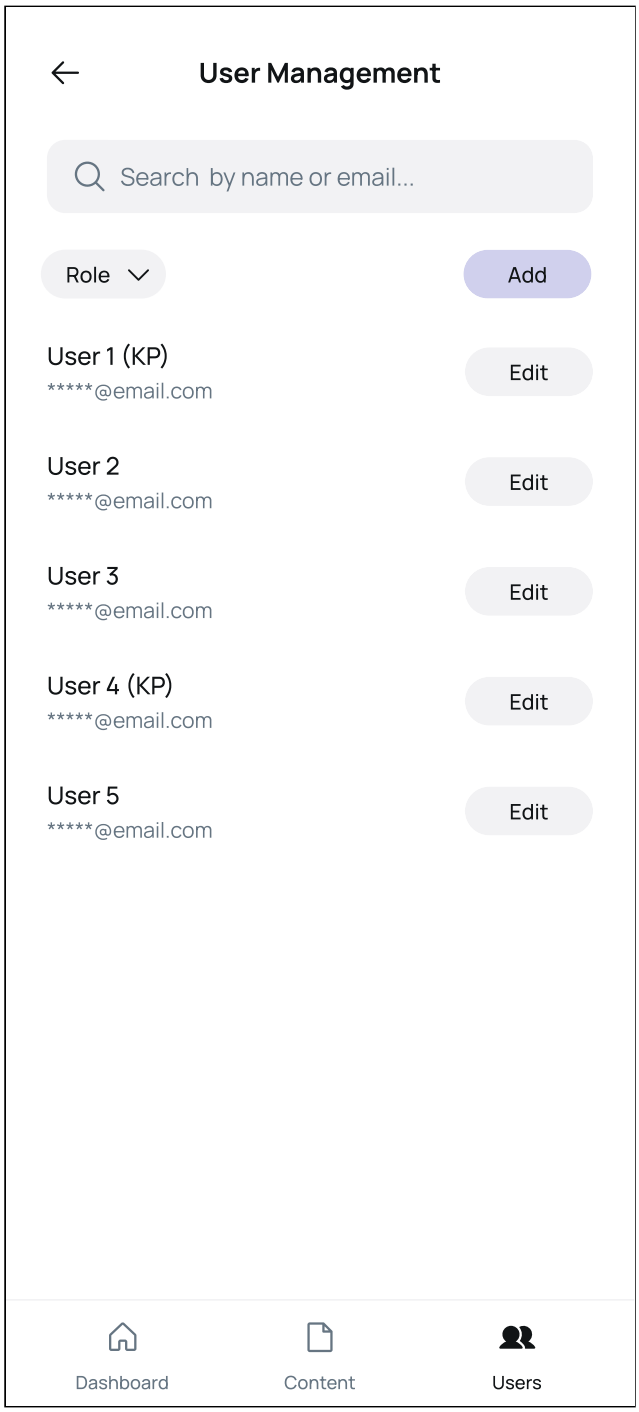

Figure 4.21: User Management

CHAPTER 5

TESTING AND IMPLEMENTATION

5.1 IMPLEMENTATION

The implementation phase involved translating the system design into a functional application. The platform is built with a Flutter frontend and leverages Firebase for its backend services. The frontend was developed using Flutter, Google's cross-platform UI toolkit. This choice was strategic, as it allows for a single codebase to be compiled for mobile (Android and iOS) and web applications. This significantly accelerates development and ensures a consistent user experience across all devices. The UI was built using a modular widget-based architecture, making it scalable and easy to maintain. For the backend, Firebase was chosen for its comprehensive suite of managed services. Firestore, a NoSQL document database, was used to store all application data, such as user profiles, subjects, and student progress, due to its real-time synchronization capabilities and flexible data model. Firebase Authentication was integrated to provide a secure and easy-to-use system for managing user login, registration, and role-based access control. All AI-driven functionalities, such as quiz generation and tutoring, were handled by making secure calls to the Google Gemini API through a lightweight server-side proxy.

5.2 TESTING

A multi-layered testing strategy was employed throughout the development lifecycle. System testing was aligned with the four agile sprints defined in the project plan. The following tables detail the end-to-end test cases executed, covering both backend functionality and user interface (UI) validation.

5.2.1 Test Case 1

Focus: Core user roles, authentication, and initial administrative setup.

No.	**Date**	**Action**	**Expected Result**	**Actual Result**	**Pass? (Yes)**
1	12/08/25	A user logs in with valid credentials on the Login Screen.	User is authenticated via Firebase Auth and redirected to their role-specific dashboard.	User redirected to correct dashboard.	Yes
2	12/08/25	An Admin adds a new user (KP or Admin) via the User Management Screen.	A new user account is created in Firebase, and the user appears in the UI list with the correct role.	User appeared in list as expected.	Yes
3	19/08/25	An Admin creates a new subject and assigns a Knowledge Provider on the Create New Subject Screen.	The subject is created in Firestore and becomes visible on the assigned KP's dashboard.	Subject appeared on KP dashboard.	Yes

4	19/08/25	A Knowledge Provider uploads a syllabus for a subject on the **Syllabus Upload Screen**.	The syllabus is processed, and the AI generates a structured list of topics for review.	AI-generated topic list was displayed.	Yes

Table 5.1: Test Case 1

5.2.2 Test Case 2

Focus: Content management workflow for Knowledge Providers and student profile management.

No.	**Date**	**Action**	**Expected Result**	**Actual Result**	**Pass? (Yes)**
5	08/09/25	A KP marks modules for AI content generation from the **Upload Material Screen**.	The system correctly identifies which modules require synthetic content generation versus material-based generation.	System flagged correct modules.	Yes
6	26/08/25	A KP reviews and accepts the AI-generated content for a topic on the **Review Content Screen**.	The topic's status in Firestore is updated from 'generated' to 'pending_review'.	Firestore status updated as expected.	Yes

7	26/08/25	A student updates their availability and weekly schedule on the **User Profile Management Screen**.	The changes are saved to the 'availability' map in the user's Firestore document.	Data saved to Firestore successfully.	Yes

Table 5.2: Test Case 2

5.2.3 Test Case 3

Focus: Core student learning loop, from scheduling to initial assessment.

No.	Date	Action	Expected Result	Actual Result	Pass? (Yes)
8	16/09/25	A student enrolls in a new subject from the **Subject List Screen**.	A new document is created in 'student_schedules' and an AI-generated schedule is populated.	Schedule was created as expected.	Yes
9	16/09/25	A student opens a scheduled topic and views the material on the **Learning Content Viewer**.	The AI-personalized learning segments, including text and examples, are displayed correctly.	Content viewer loaded correct data.	Yes

10	23/09/25	A student interacts with the AI tutor by asking a question related to the content.	The AI tutor provides a relevant, context-aware response based on the current topic.	AI response was relevant and helpful.	Yes
11	23/09/25	A student completes a learning segment and takes the **AI-Generated Quiz**.	A quiz with relevant multiple-choice questions is presented to the student.	Relevant quiz was generated.	Yes

Table 5.3: Test Case 3

5.2.4 Test Case 4

Focus: Completing the adaptive feedback cycle and visualizing student progress.

No.	**Date**	**Action**	**Expected Result**	**Actual Result**	**Pass? (Yes)**
12	30/09/25	A student submits a quiz and views the results.	The score is calculated, and instant feedback with explanations for incorrect answers is provided.	Score and explanations displayed.	Yes
13	30/09/25	The system checks for a student's missed topics and adjusts the schedule.	Any overdue, incomplete topics are automatically rescheduled for a future date.	Schedule was updated as expected.	Yes

14	07/10/25	A student navigates to the **Progress Tracking Dashboard** to view their performance.	The dashboard accurately visualizes overall progress, completed topics, and quiz scores.	Dashboard visuals matched user data.	Yes

Table 5.4: Test Case 4

CHAPTER 6

RESULTS AND DISCUSSION

6.1 SYSTEM FUNCTIONALITY AND PERFORMANCE

The developed prototype successfully integrates the core functionalities for all three user roles: Student, Knowledge Provider, and Administrator. End-to-end testing confirms that the workflows are operational and the system behaves as designed.

- **Student Experience:** Students can successfully register, enroll in subjects, and receive an AI-generated personalized study schedule. They can navigate through the learning content, interact with the AI tutor, and test their knowledge with on-the-fly quizzes. The progress tracking dashboard provides a clear visual summary of their performance.

- **Knowledge Provider Workflow:** The interface allows Knowledge Providers to upload educational materials in PDF and TXT format. The system's backend successfully processes these documents, leveraging the AI to structure the content into digestible learning modules, significantly reducing the manual effort required for course creation.

- **Administrative Oversight:** The administrator dashboard provides essential tools for managing users, creating new subjects, and reviewing and approving the AI-processed content. This human-in-the-loop approach ensures the quality and accuracy of the learning materials presented to students.

6.2 EVALUATION OF AI-POWERED FEATURES

The central hypothesis of this project was that a Large Language Model could be leveraged to create a truly adaptive learning experience. The results largely validate this hypothesis.

- **Personalized Learning Path:** The Gemini API proved highly effective at generating coherent and realistic study schedules based on user inputs. The schedules correctly allocated time for different topics and were logically sequenced.
- **AI-Generated Quizzes:** The quality of the quizzes generated by the AI was consistently high. The questions were relevant to the preceding content, and the inclusion of explanations for correct answers provided immediate, valuable feedback to the learner, reinforcing the concepts.
- **AI Tutor:** The conversational AI tutor, powered by the same LLM, was capable of answering a wide range of student queries in a context-aware manner. It could simplify complex topics, provide alternative examples, and guide students without simply giving away the answer.

6.3 CHALLENGES AND LIMITATIONS

Despite the overall success, the project faced several challenges and has inherent limitations.

- **API Dependency:** The system's core functionality is entirely dependent on the availability and performance of the external Google Gemini API. Any latency, downtime, or change in the API's pricing or terms of service would directly impact AuraLearn.
- **Content Accuracy (Hallucinations):** While infrequent, there were instances during testing where the LLM would "hallucinate" or generate factually incorrect

information. This underscores the critical importance of the administrator review step in the content pipeline to ensure the academic integrity of the platform.

- **Limited Scope:** The current prototype focuses on text-based content from PDF and TXT files. It does not support other media types like videos, interactive diagrams, or complex mathematical equations, which limits its applicability for certain subjects.

- Cost Considerations: Extensive use of a powerful LLM like Gemini incurs costs based on token usage. A full-scale deployment of AuraLearn would require a robust cost management and optimization strategy.

6.4 ACHIEVEMENT OF OBJECTIVES

The project successfully met all the primary objectives set out at its inception.

- The system develops a dynamic, personalized learning path, fulfilling the first objective.

- A context-aware AI tutoring system was implemented, meeting the second objective.

- An AI-assisted content ingestion workflow was created for Knowledge Providers, achieving the third objective.

- The platform integrates on-the-fly, AI-generated assessments, satisfying the fourth objective.

- A functional prototype with distinct user roles was built, successfully delivering on the final objective.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This project successfully demonstrated the potential of modern Large Language Models to create a scalable, personalized learning platform, AuraLearn. By integrating the capabilities of the Google Gemini 2.5 Flash API, AuraLearn delivers a dynamic learning experience that adapts to individual student needs, a task that was previously complex and resource-intensive. The final prototype provides personalized schedules, context-aware tutoring, and instant, formative assessments, directly addressing the core limitations of the traditional one-size-fits-all educational model. The project has confirmed that LLMs can serve as a powerful engine for creating next-generation educational tools. While challenges such as API dependency and the need for human oversight remain, AuraLearn stands as a successful validation of the core concept and provides a solid foundation for future development in the field of AI-driven education.

7.2 FUTURE WORK

While the current version of AuraLearn is a robust prototype, there are numerous avenues for future enhancement and research:

- **Advanced Student Modeling:** Develop a more sophisticated model of student knowledge that tracks mastery of individual concepts over time. This would allow the AI to recommend remedial topics more accurately and adapt the difficulty of content and quizzes with greater precision.

- **Multi-modal Content Support:** Extend the content ingestion pipeline to handle a variety of media formats, including video transcripts, audio lectures, and images, to create a richer and more engaging learning environment.

- **Gamification and Engagement Analytics:** Incorporate gamification elements such as points, badges, and leaderboards to increase student motivation. Furthermore, implement analytics to track user engagement and identify which content formats are most effective].

- **Enhanced RAG Implementation:** Improve the Retrieval-Augmented Generation (RAG) system for the AI tutor by using more advanced vector search techniques. This would allow the tutor to provide answers that are more strictly grounded in the uploaded course material, reducing the risk of hallucinations.

- **Collaborative Learning Features:** Introduce features that facilitate peer-to-peer learning, such as AI-moderated discussion forums or collaborative problem-solving spaces.

- **Accessibility and Internationalization:** Enhance the platform to meet web accessibility standards (WCAG) and add support for multiple languages to make the tool globally accessible.

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[1] B. S. Bloom, "The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring," *Educational Researcher*, vol. 13, no. 6, pp. 4–16, 1984.

[2] P. Brusilovsky, "Adaptive hypermedia," *User modeling and user-adapted interaction*, vol. 11, no. 1-2, pp. 87–110, 2001.

[3] A. C. Graesser, M. W. Conley, and A. M. Olney, "Intelligent tutoring systems," in *Handbook of educational psychology*, L. Corno and E. M. Anderman, Eds. New York, NY: Routledge, 2016, pp. 301–314.

[4] M. Heilman, "Automatic factual question generation from text," Ph.D. dissertation, Carnegie Mellon University, 2010.

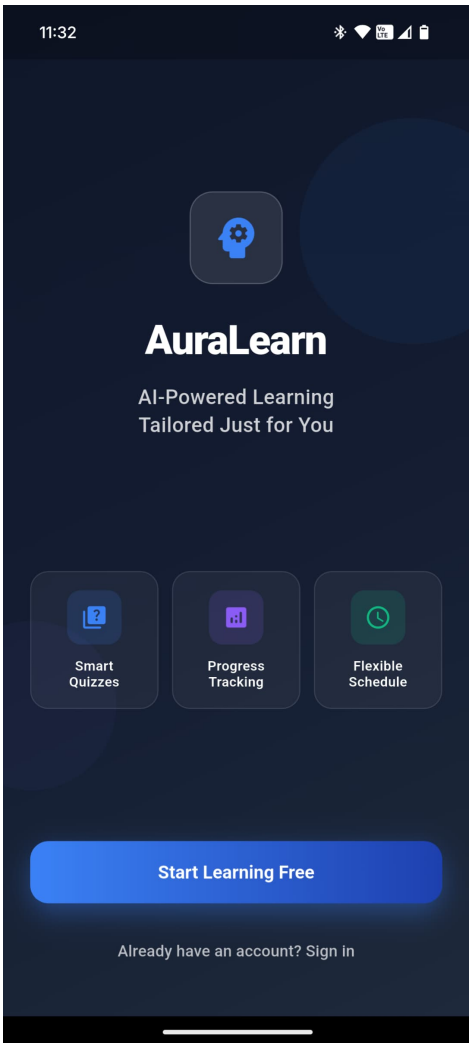
[5] Gemini Team, Google, "Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities," arXiv preprint arXiv:2507.06261, 2025. [Online]. Available: `https://arxiv.org/abs/2507.06261`

APPENDICES

SCREENSHOTS

Landing Screen

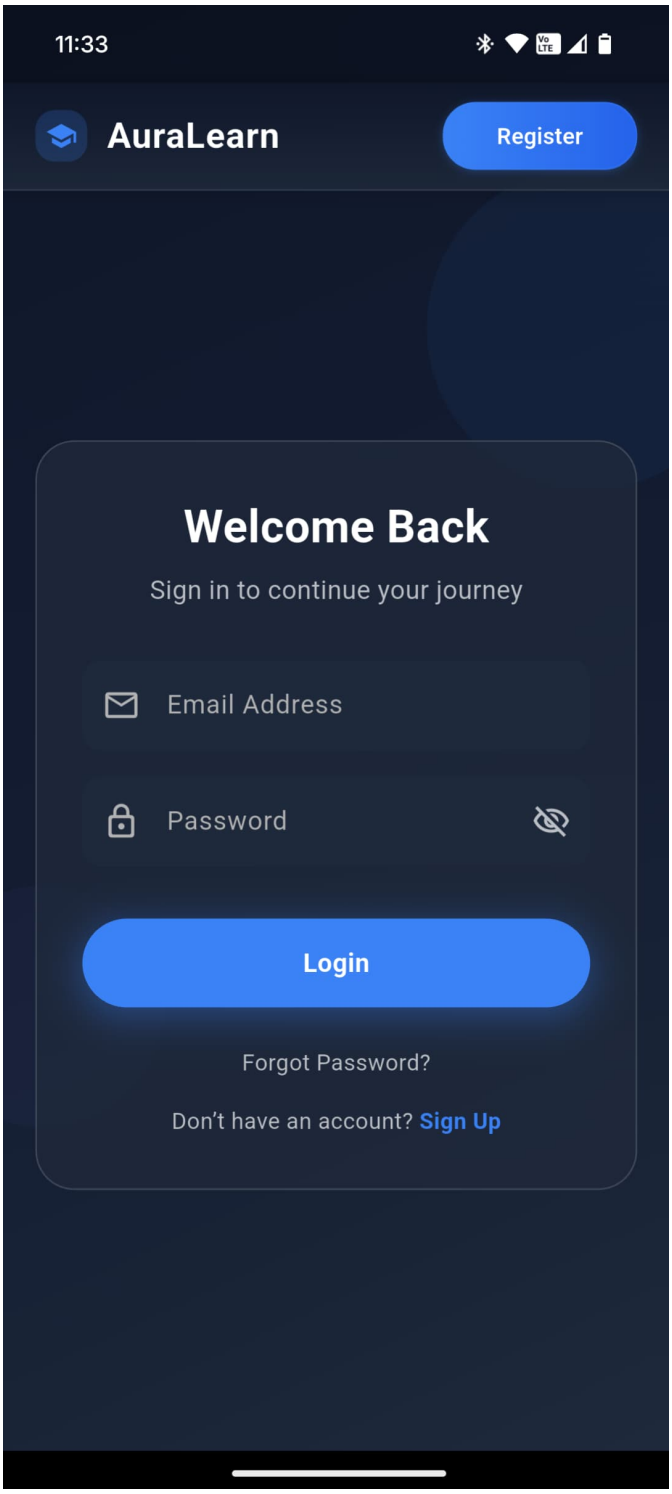
The initial screen that introduces the app's features and prompts users to start learning or sign in.



Application Landing Screen

Login Screen

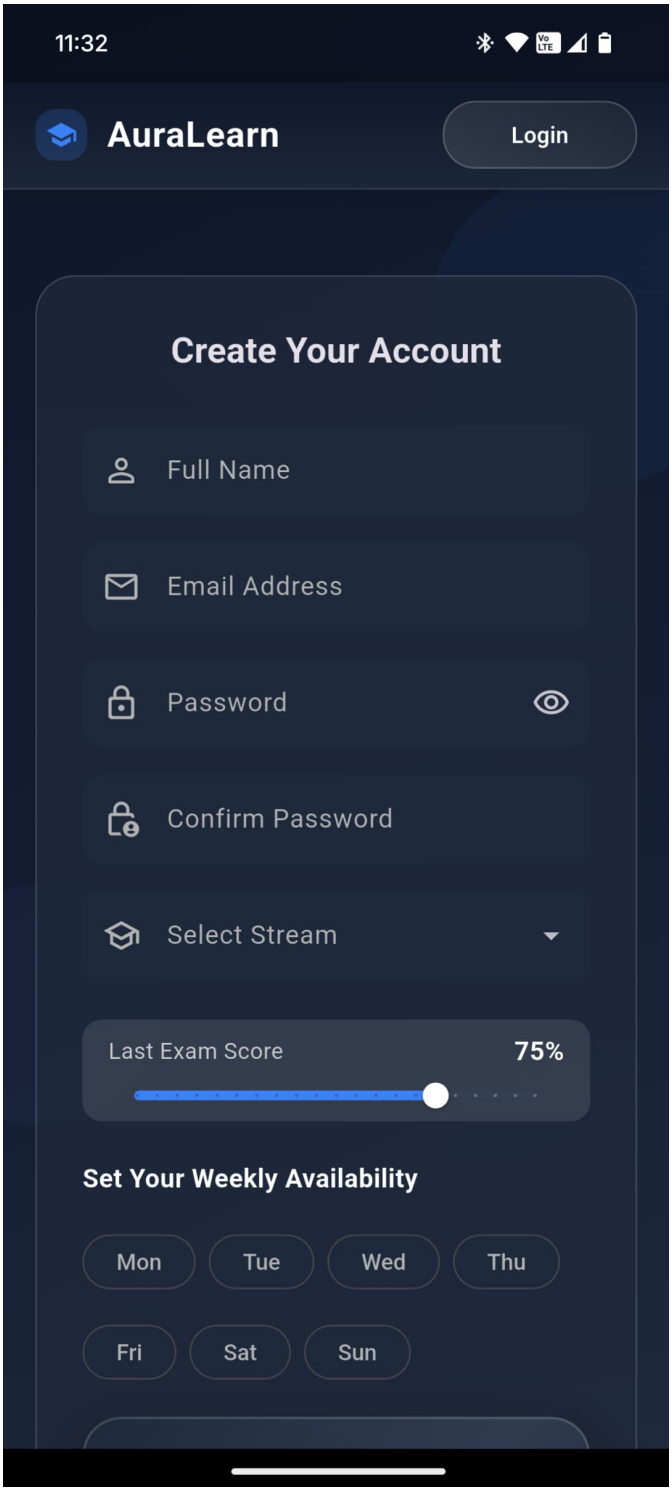
A secure interface for existing users to enter their credentials and access their accounts.



User Login Screen

Student Registration

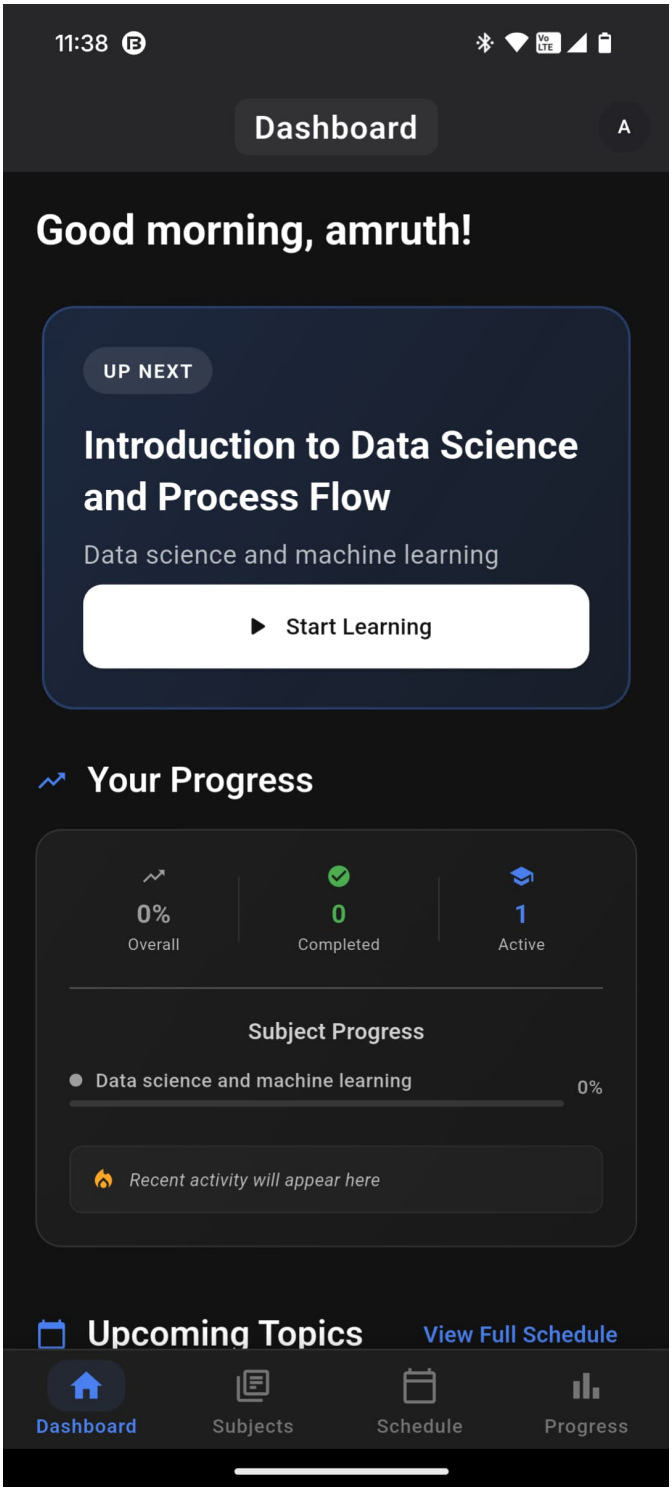
A form for new students to create an account, providing details like their stream and weekly availability.



Student Registration Screen

Student Dashboard

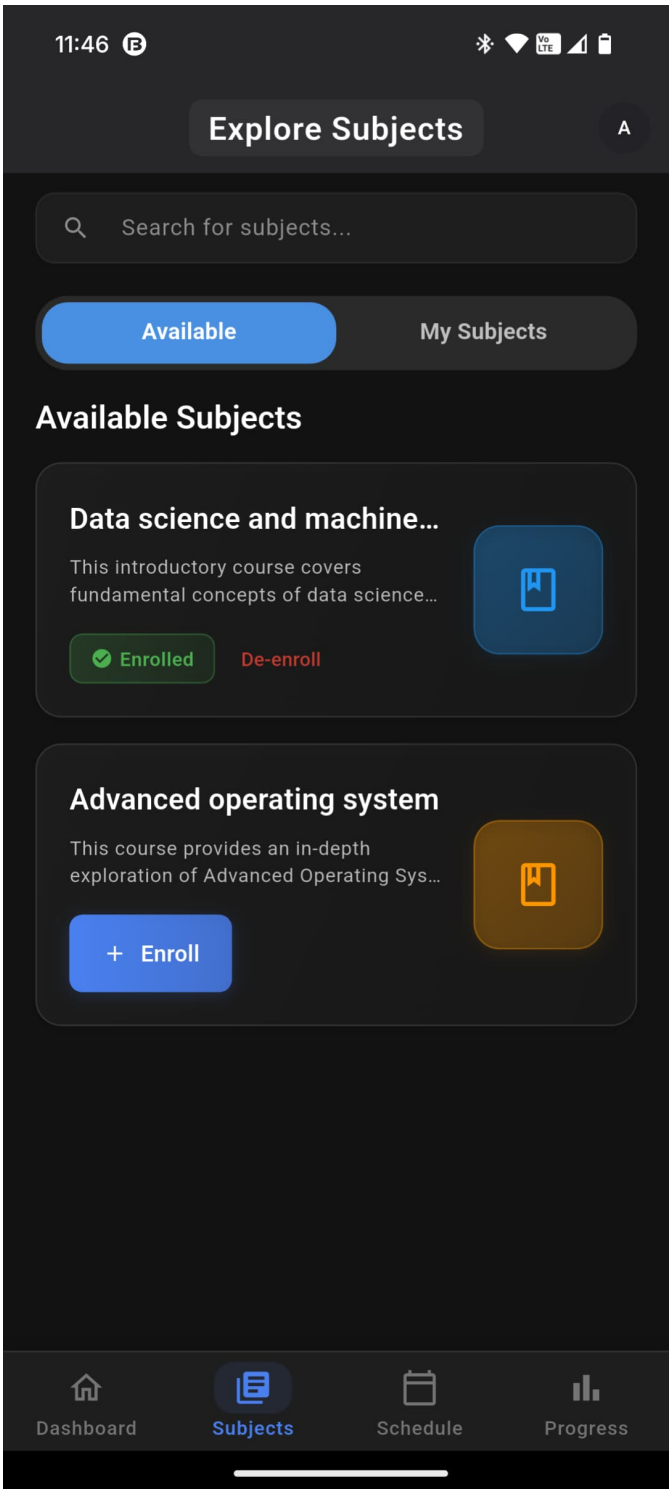
The main hub for students, showing their next scheduled topic and an overview of their progress.



Student Dashboard Screen

Subject Exploration Screen

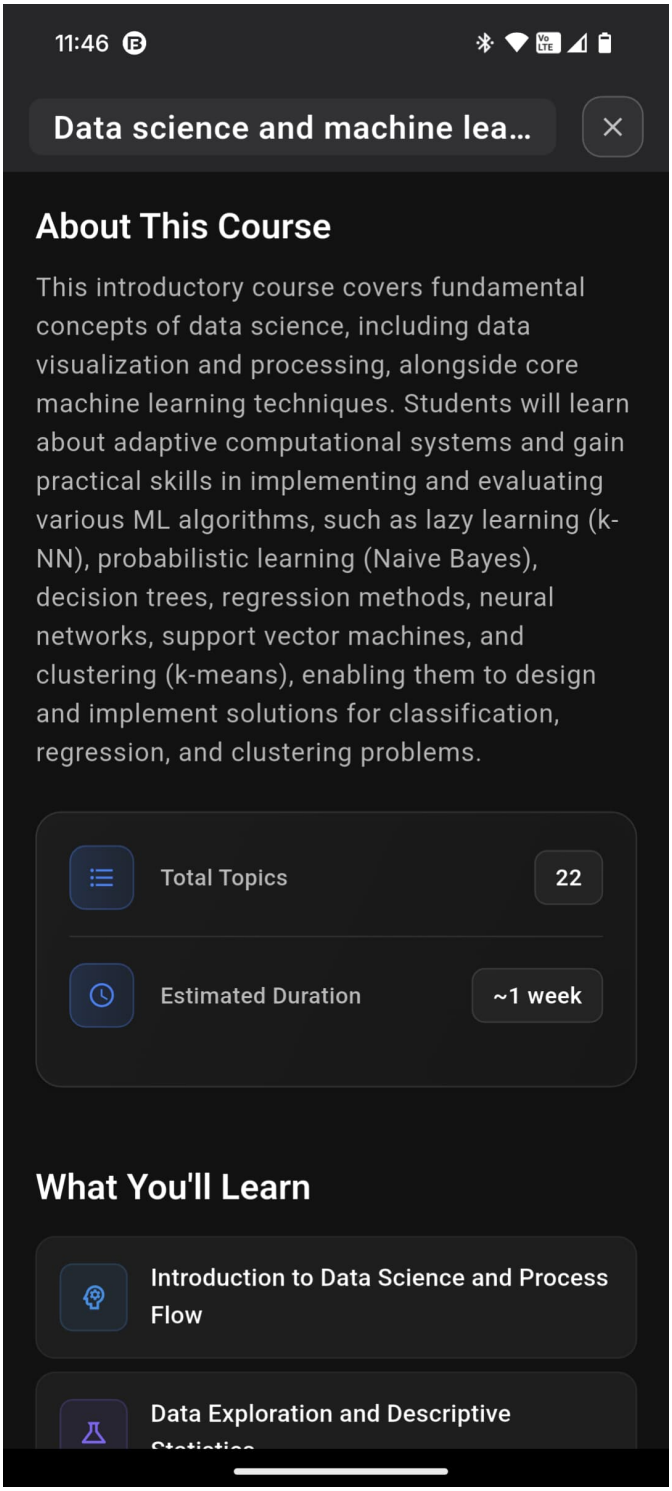
Allows students to browse and search for available subjects and enroll in new courses.



Subject Exploration and Enrollment

Subject Details

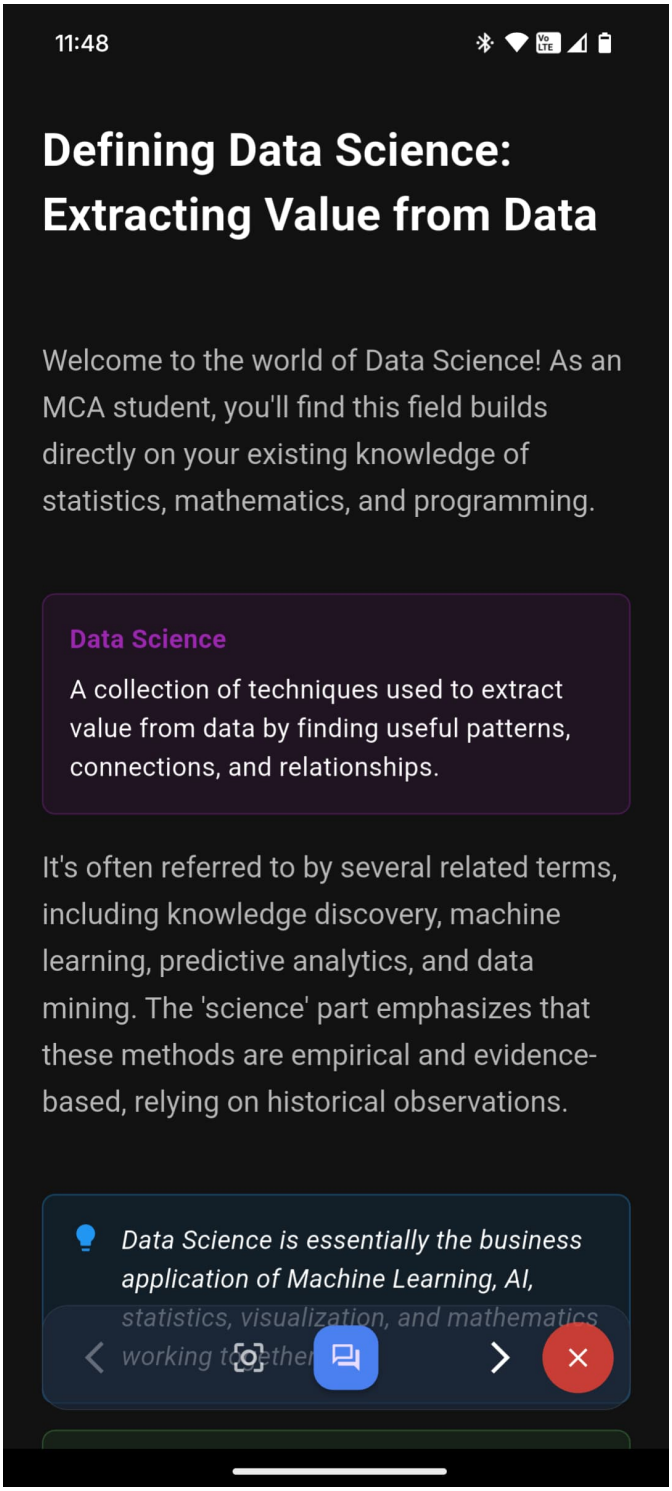
Provides detailed information about a course, including total topics and estimated duration, before enrollment.



Subject Details Screen

Learning Content Viewer

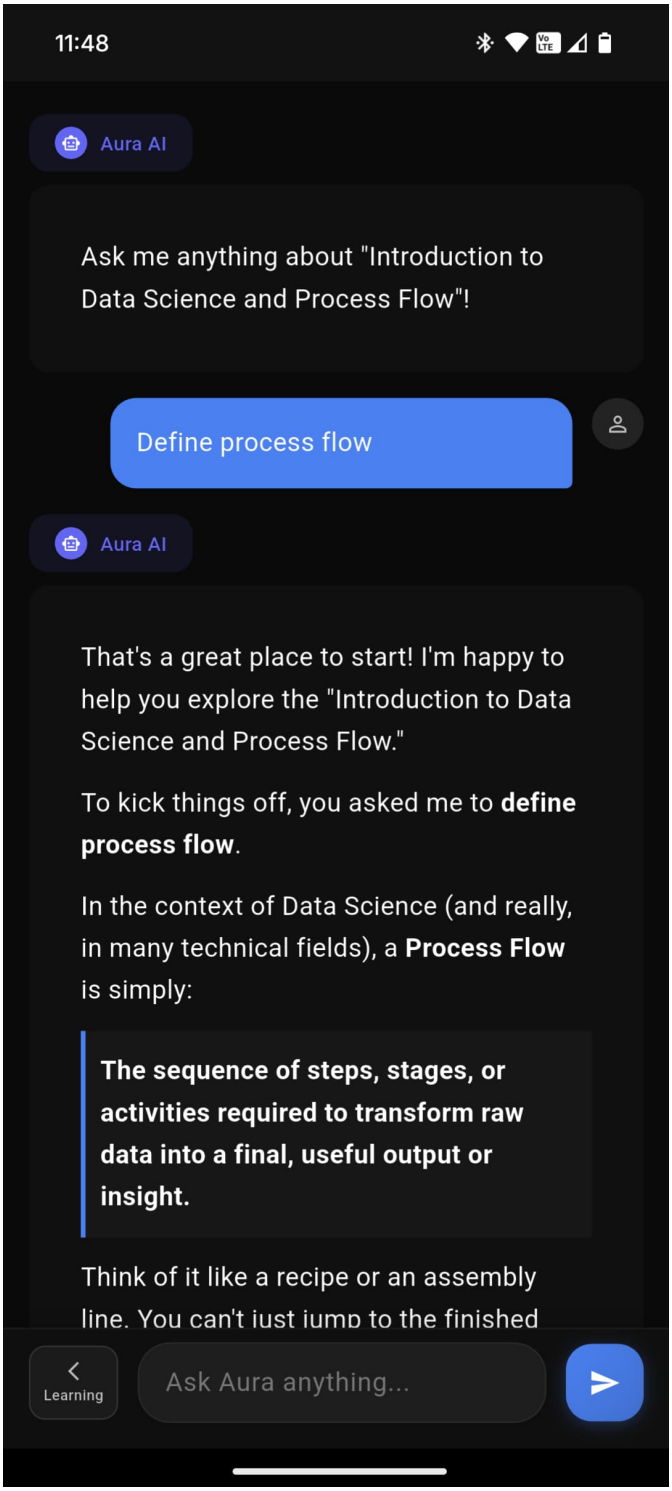
The screen where students read the AI-processed content chunks for a specific topic.



Learning Content Viewer

AI Tutor Interface

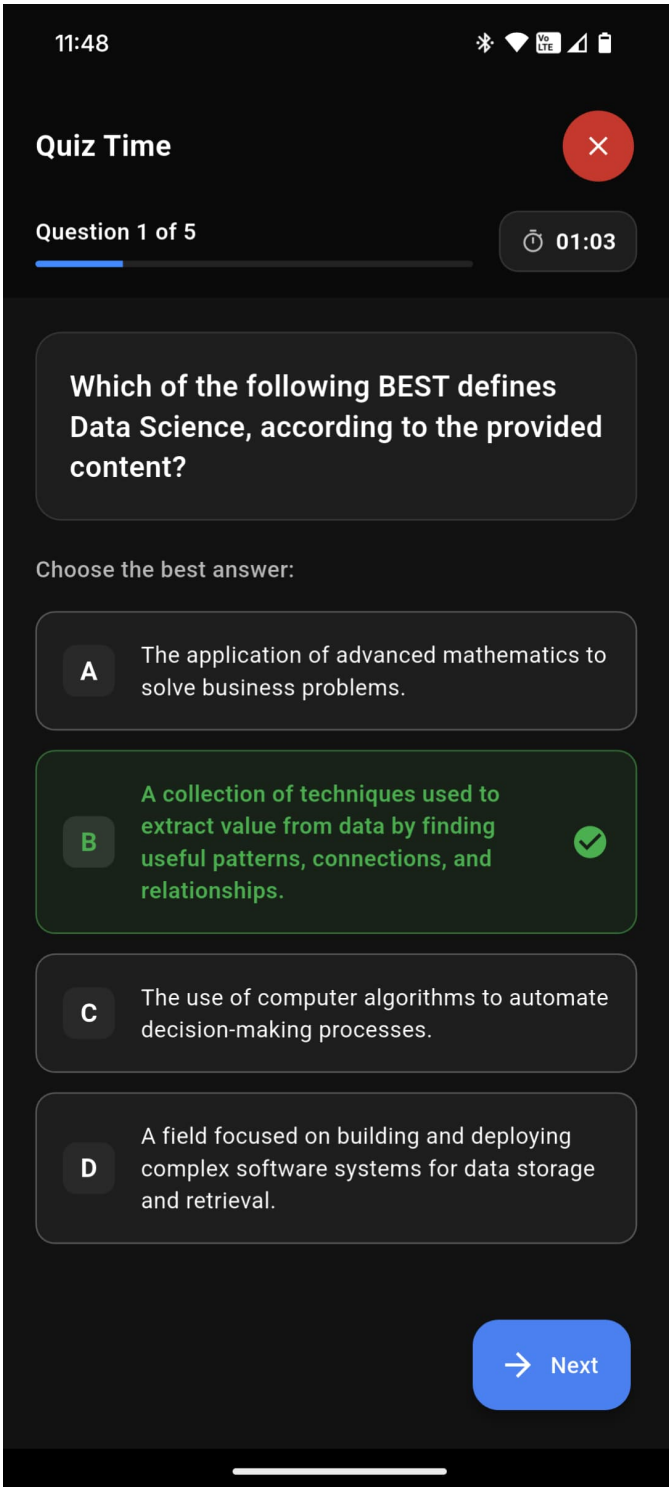
A conversational chat interface where students can ask questions and get real-time, context-aware help from the AI.



AI Tutor Chat Screen

AI-Generated Quiz

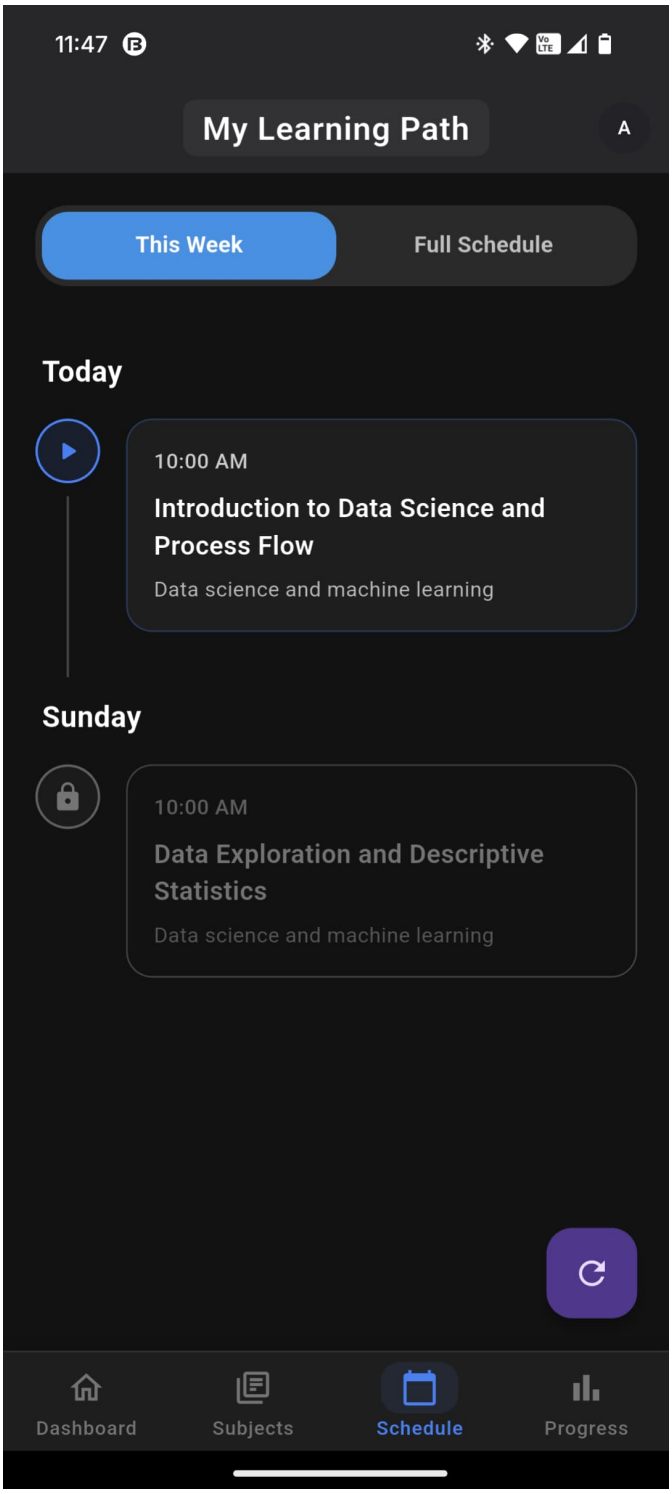
An on-the-fly quiz generated by the AI to test a student's understanding of the content they just studied.



AI-Generated Quiz Screen

Student Schedule View

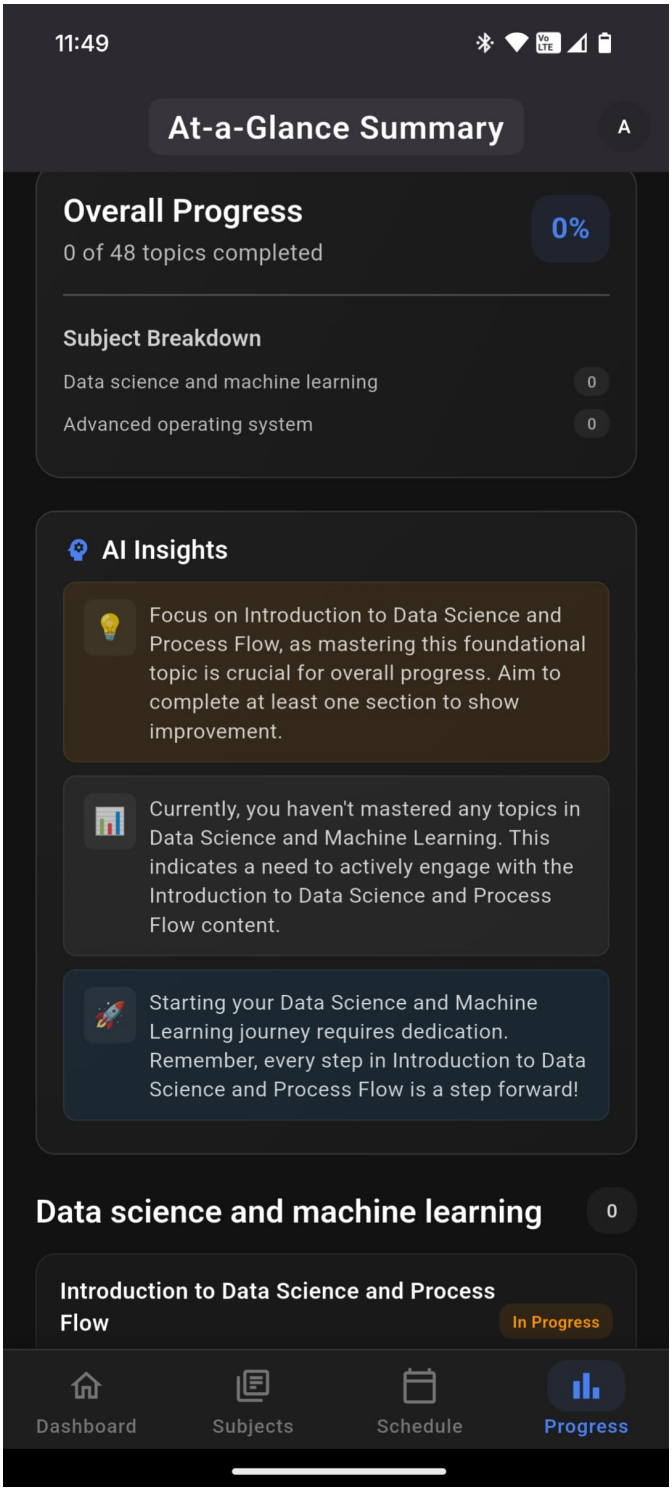
Displays the student's personalized learning path and scheduled topics for the current week.



Student Weekly Schedule

Progress Tracking Dashboard

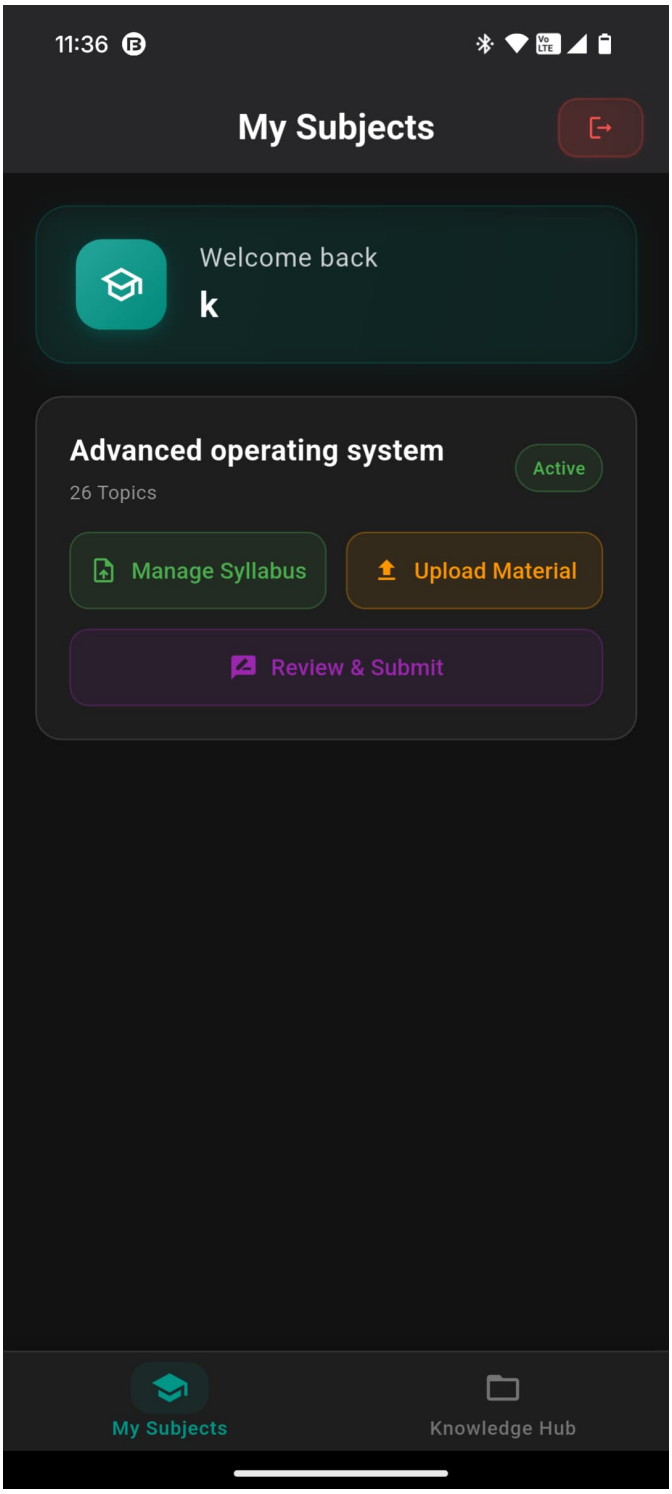
Visualizes a student's overall progress, subject completion, and provides AI insights on where to focus.



Student Progress Dashboard

Knowledge Provider Dashboard

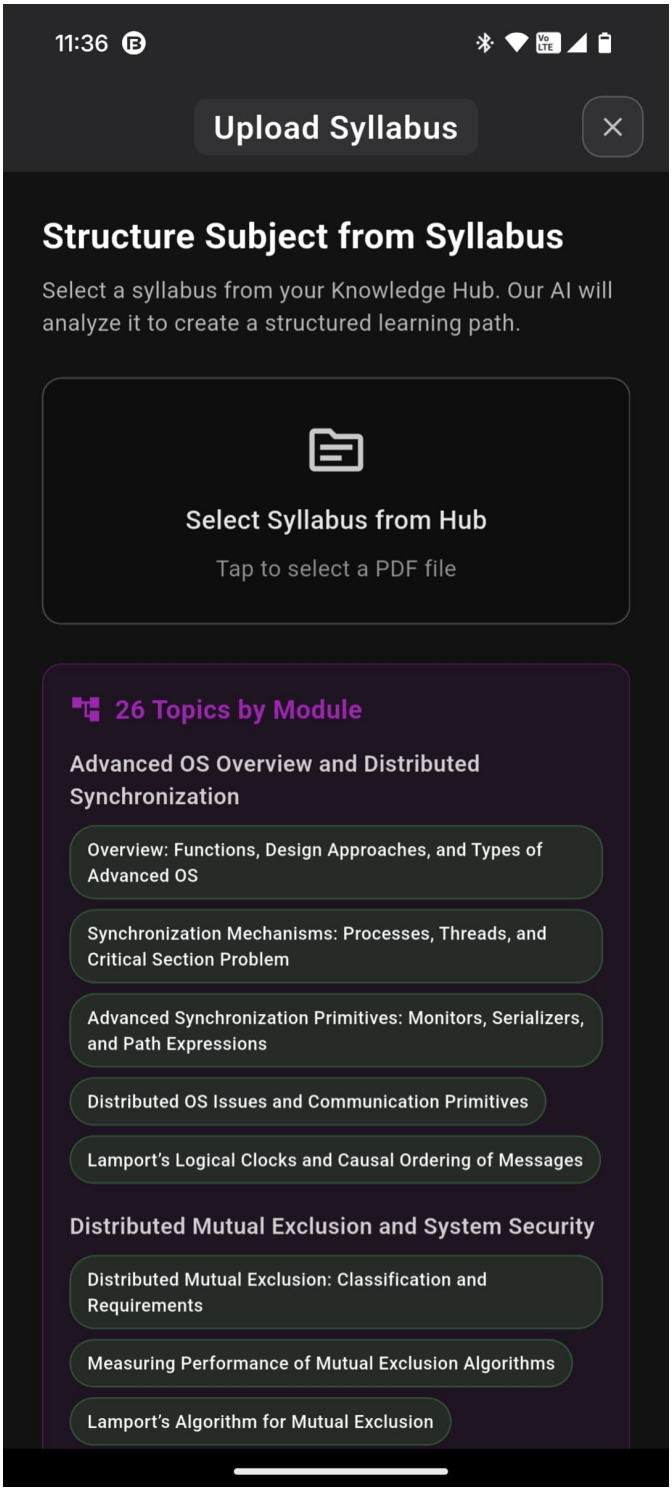
The main dashboard for KPs to manage the subjects they are responsible for and track content status.



Knowledge Provider Dashboard Screen

Syllabus Upload Interface

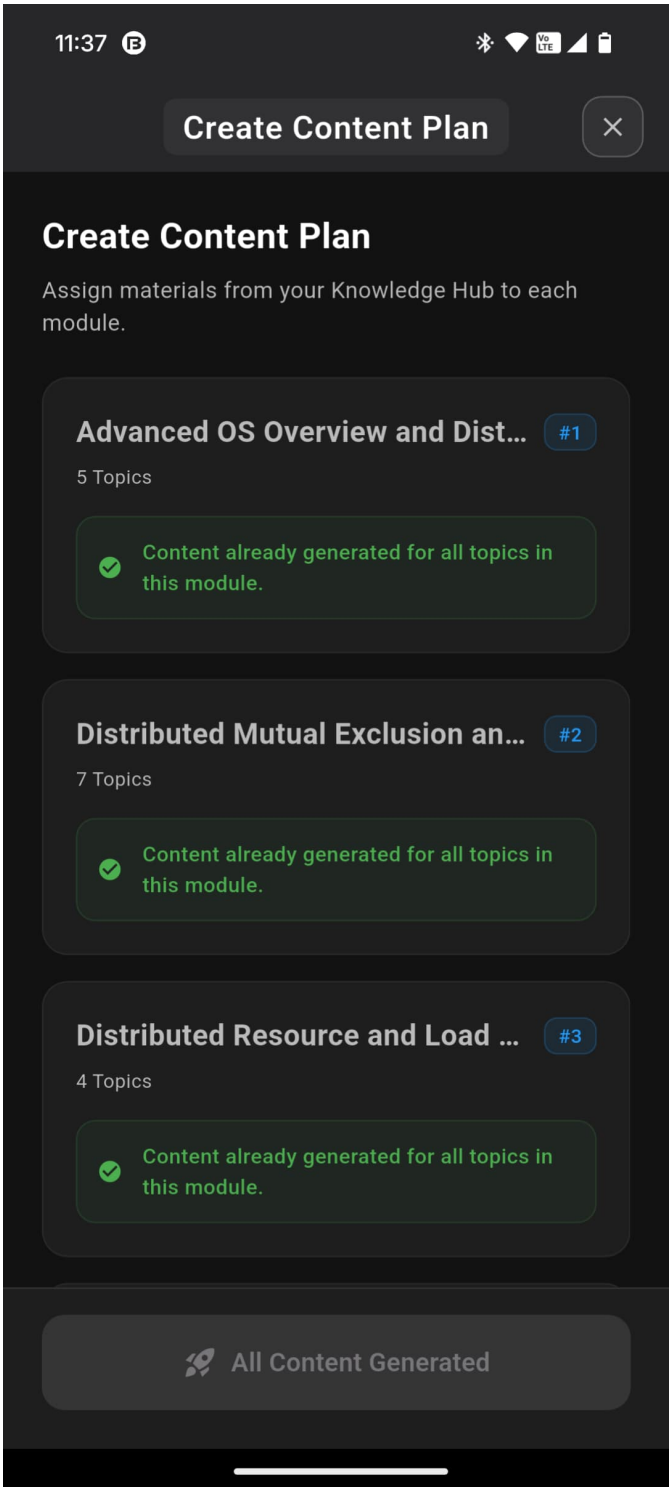
Allows Knowledge Providers to upload a syllabus PDF, which the AI analyzes to generate a structured topic list.



Syllabus Upload and Processing Screen

Content Plan Management

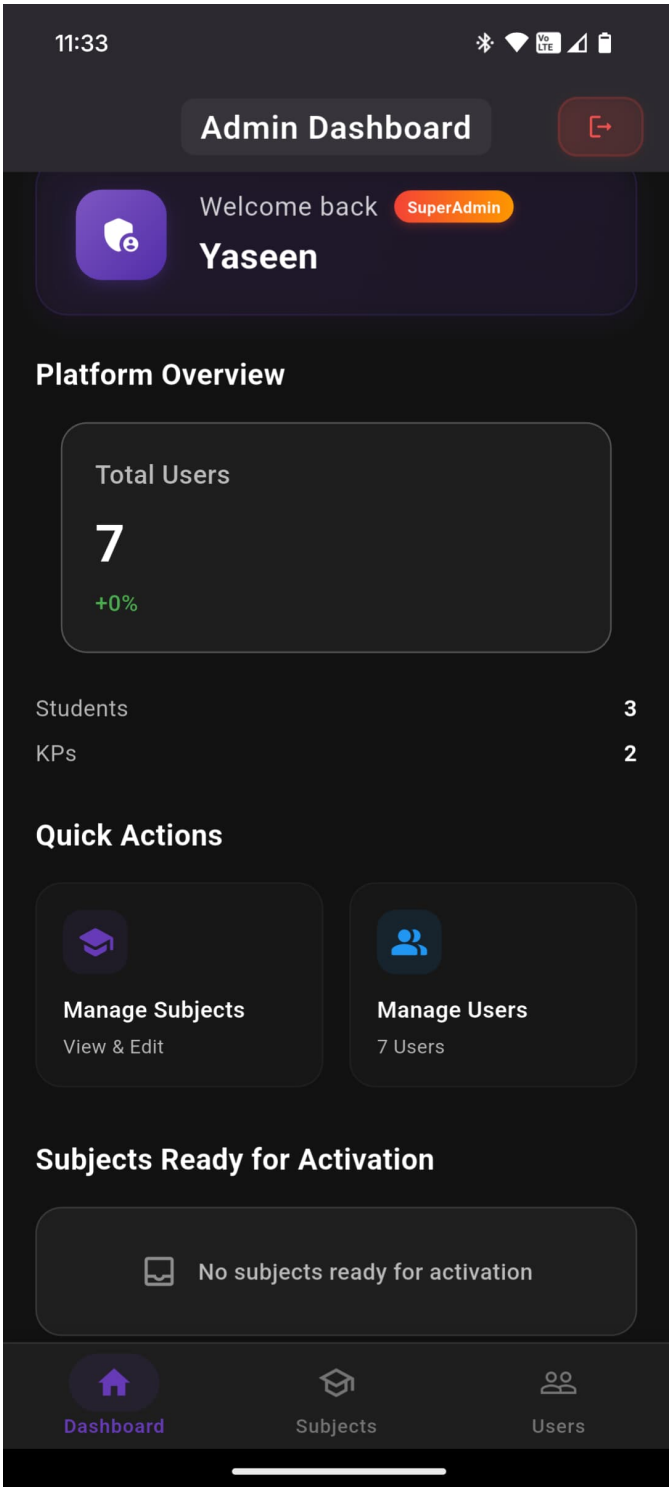
Interface for KPs to review the content generation status for different modules within a subject.



Content Plan Management Screen

Admin Dashboard

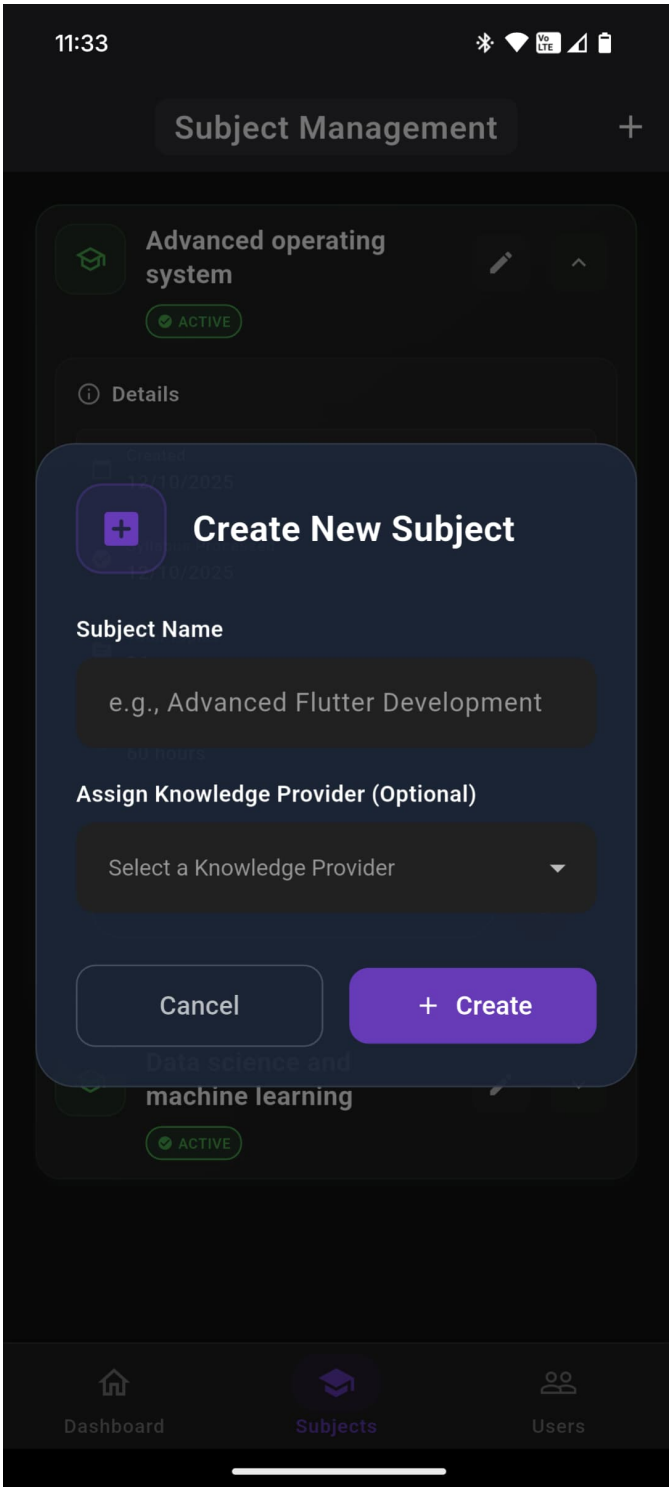
The central dashboard for administrators, providing an overview of platform statistics and quick actions.



Admin Dashboard Screen

Create New Subject

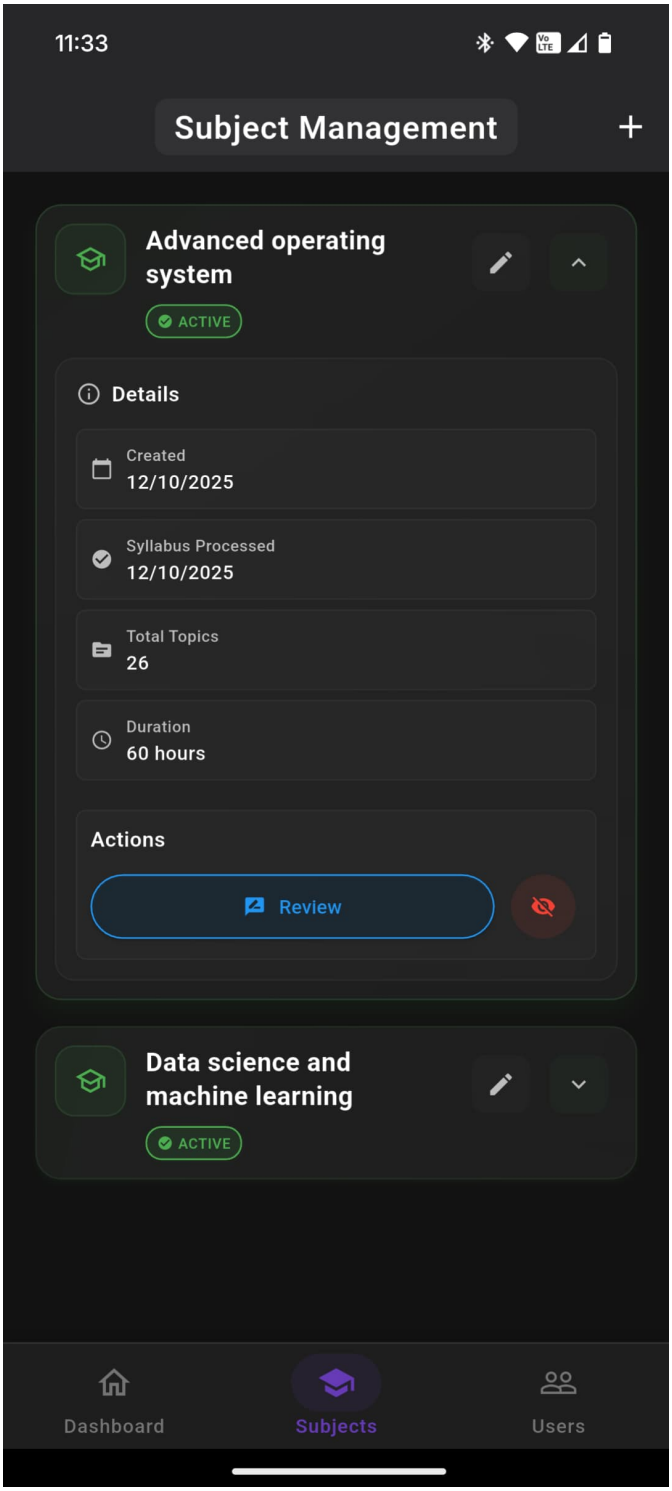
A modal form for admins to create a new subject and optionally assign it to a Knowledge Provider.



Admin: Create New Subject

Subject Management

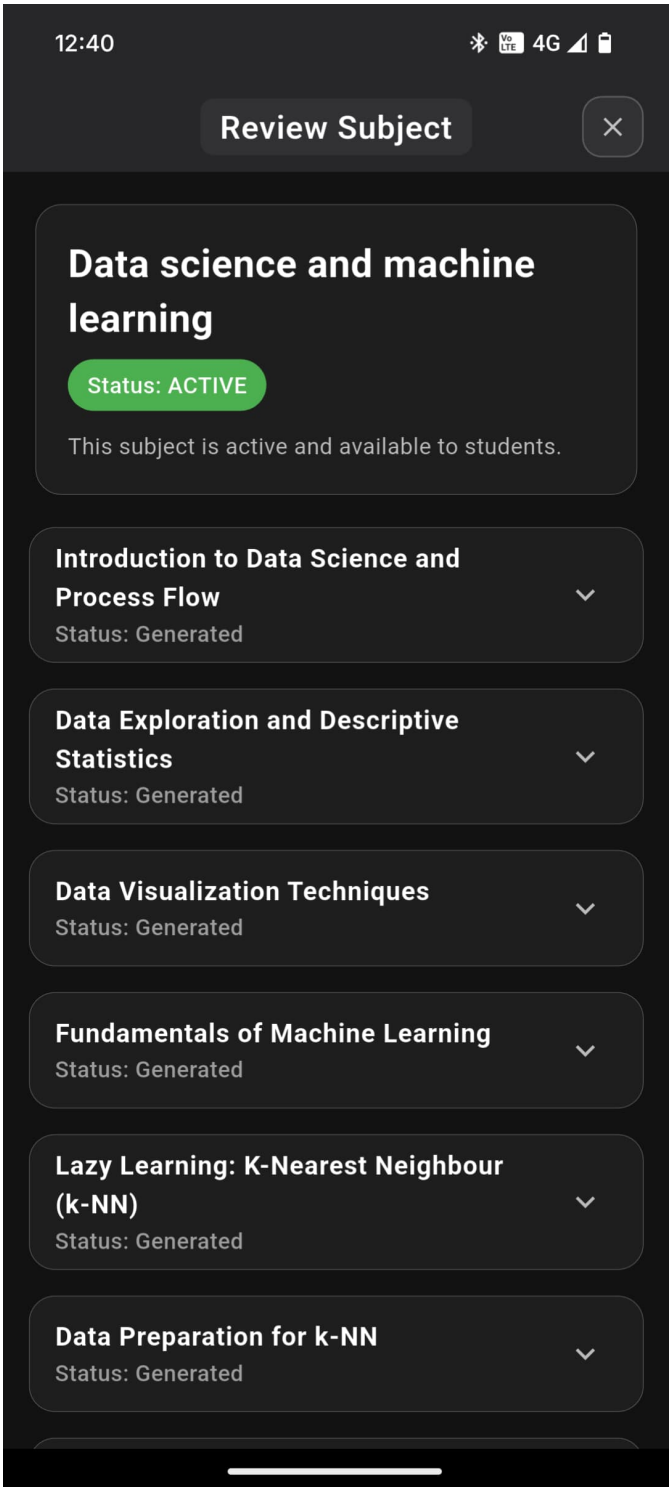
An expanded view for admins to see details, status, and management actions for a specific subject.



Admin: Subject Management

Content Review

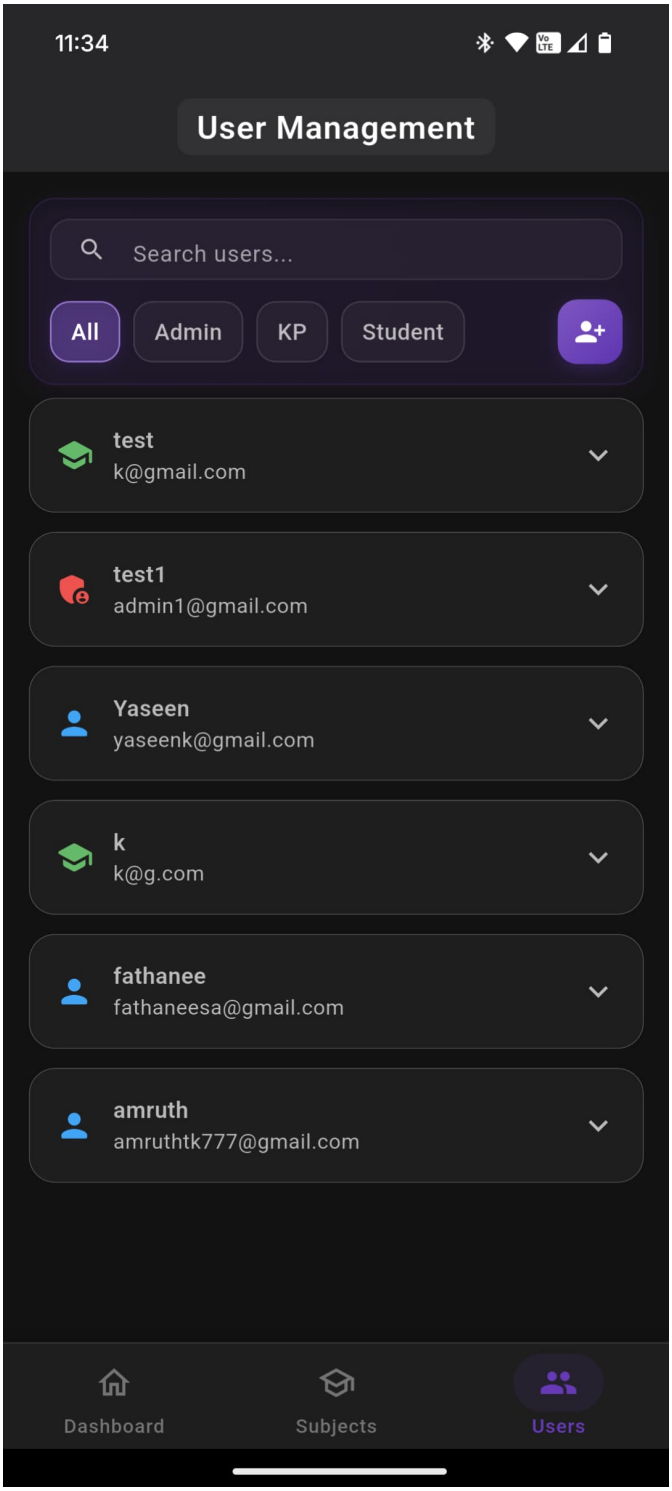
Interface for admins to review the AI-generated content for a subject before activating it for students.



Admin: Content Review Screen

User Management

A list view for admins to search, filter, and manage all user accounts and roles on the platform.



Admin: User Management Screen